Submission of project
Tittle: Secure File Storage and Access Management for Project Teams

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Objective

The objective of this project is to design and implement a **secure file storage and access management system** for finance teams at *Globex Financial*. The system focuses on enforcing **user access controls**, **role-based permissions**, **Access Control Lists (ACLs)**, **command history management**, and **security violation logging**.
The project aims to prevent unauthorized file access and modification while ensuring accountability, traceability, and compliance with audit requirements.

Implementation

To achieve the project objectives, a **Linux-based security solution** was implemented using built-in system administration and security tools. The solution was divided into multiple structured phases to ensure clarity, security, and audit readiness.

1. User and Group Management

Separate **project-based groups** were created to represent different teams (Project A and Project B). Each user was assigned to their respective group, ensuring **role-based access control (RBAC)**.
This segregation ensured that users only had access to resources relevant to their assigned project.

2. Secure Project Directory Configuration

A dedicated project directory was created to store sensitive files.

- Group ownership was assigned carefully.
- Directory permissions were configured to restrict access to authorized users only.
- The **sticky bit** was enabled to prevent users from deleting files created by others.

This setup ensured **data integrity** and **controlled collaboration**.

3. Access Control Lists (ACLs) Implementation

ACLs were configured to provide **fine-grained permission control** beyond standard Linux permissions.

- Specific users were granted read, write, and execute permissions.
- Unauthorized users and “others” were restricted from modifying or deleting files.
- Default ACLs ensured that newly created files inherited secure permissions automatically.

This prevented accidental or malicious changes to sensitive project data.

4. Command History Management

Command history limits were configured for selected users by modifying shell configuration files.

- The number of stored commands was restricted.
- This reduced exposure of sensitive command usage while still maintaining accountability.

This step enhanced **operational security** and reduced the risk of misuse.

5. Security Violation Logging Using Auditd

The Linux **auditd** service was installed and configured to monitor all read, write, execute, and attribute changes in the project directory.

- Custom audit rules were created to track access attempts.
- Logs captured both authorized and unauthorized activities.
- Audit logs were searchable using audit keys.

This provided **real-time monitoring** and **forensic evidence** for security audits.

6. Web-Based Audit Report System
To improve visibility and usability, audit logs were exported to a **web-accessible report system** using Apache.

- Logs were periodically updated using cron jobs.
- Access to audit reports was protected using HTTP authentication.

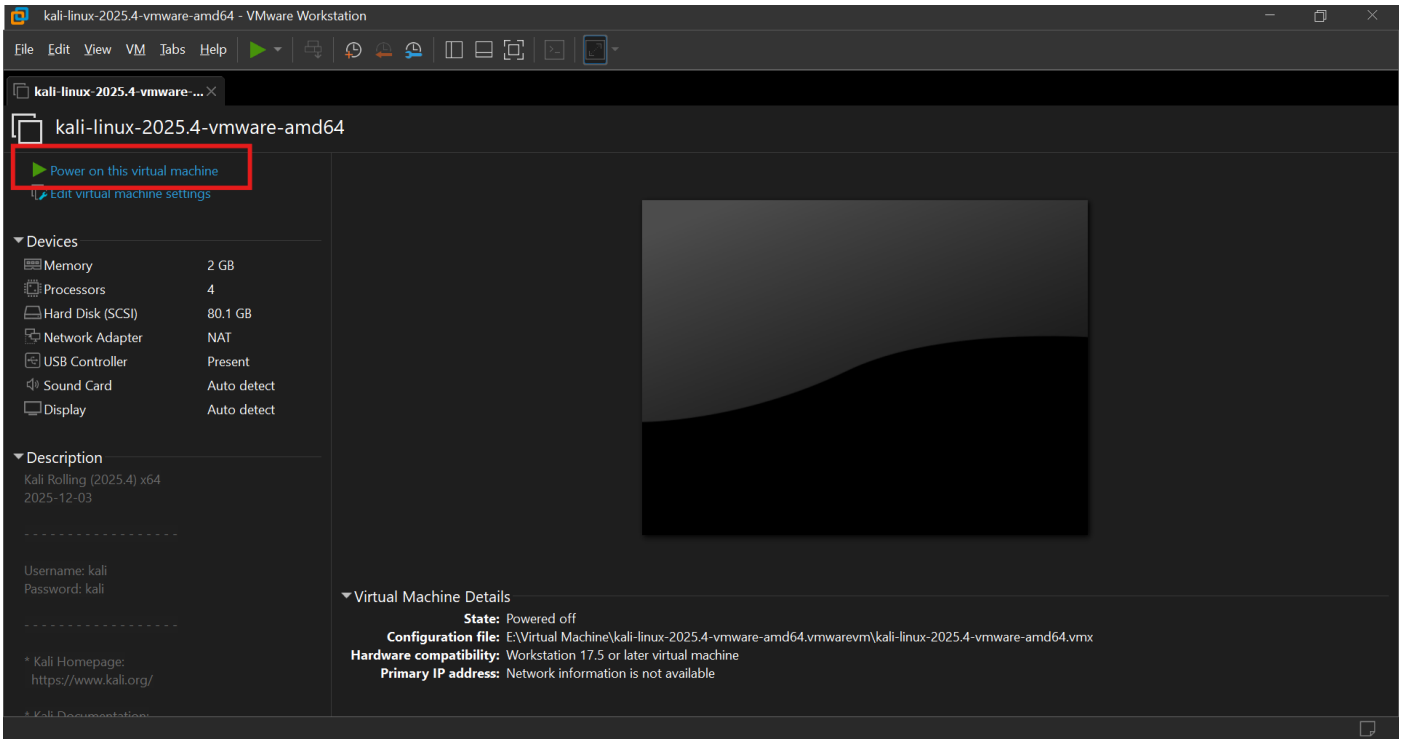
This allowed administrators to review security events easily while maintaining restricted access.

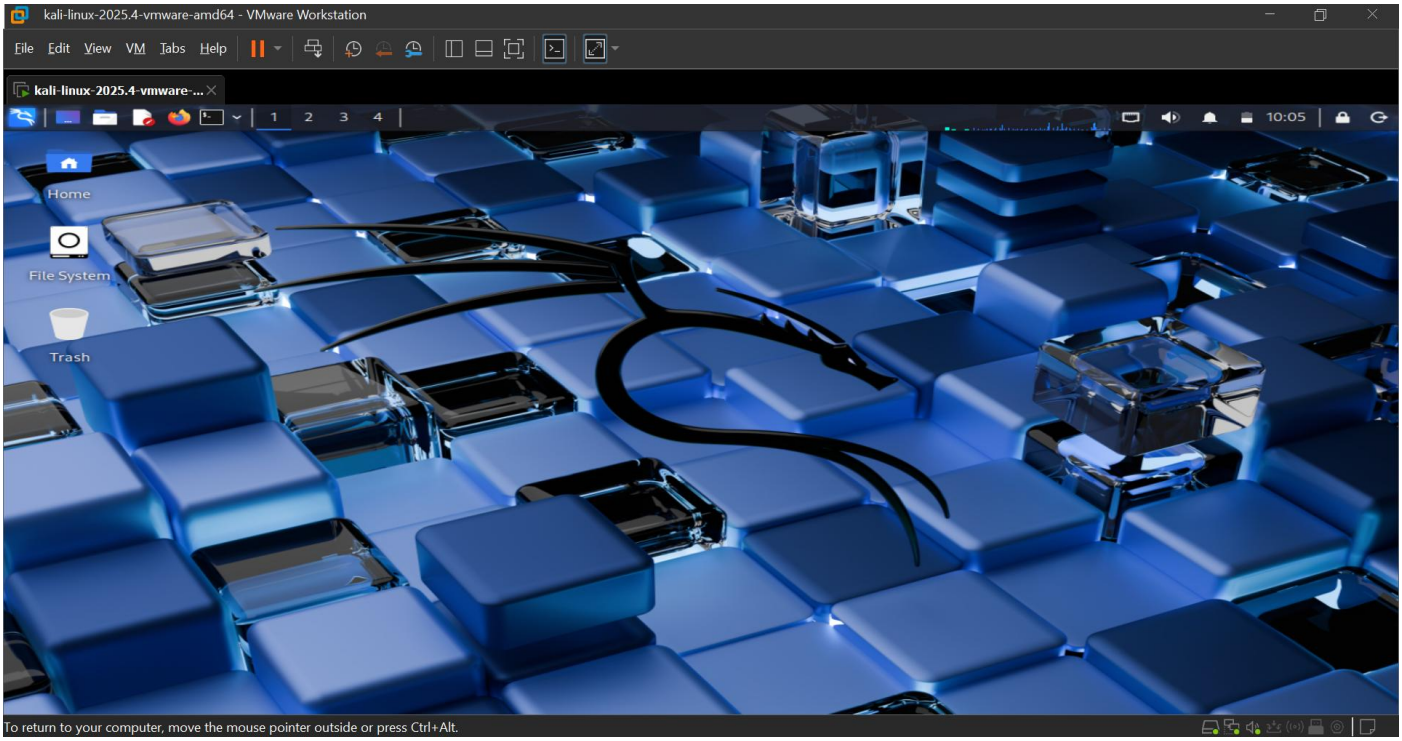
7. Testing and Validation
Multiple tests were conducted to validate the system:

- Authorized users successfully created and accessed files.
- Unauthorized users were denied access.
- File deletion restrictions worked as expected.
- Command history limits were enforced correctly.
- Security events were logged and visible in audit reports.

All tests confirmed that the system worked as intended.

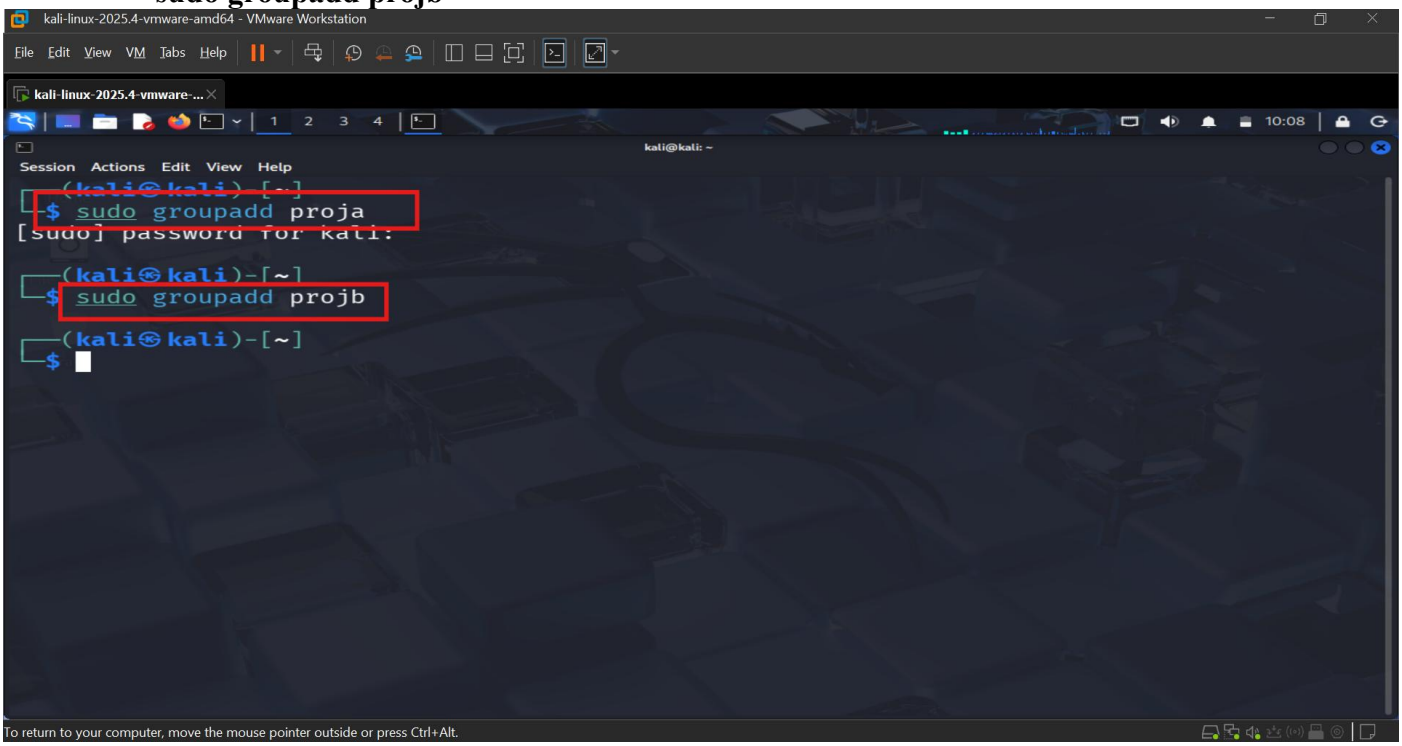
Step 1: Powering the machine and login using credentials



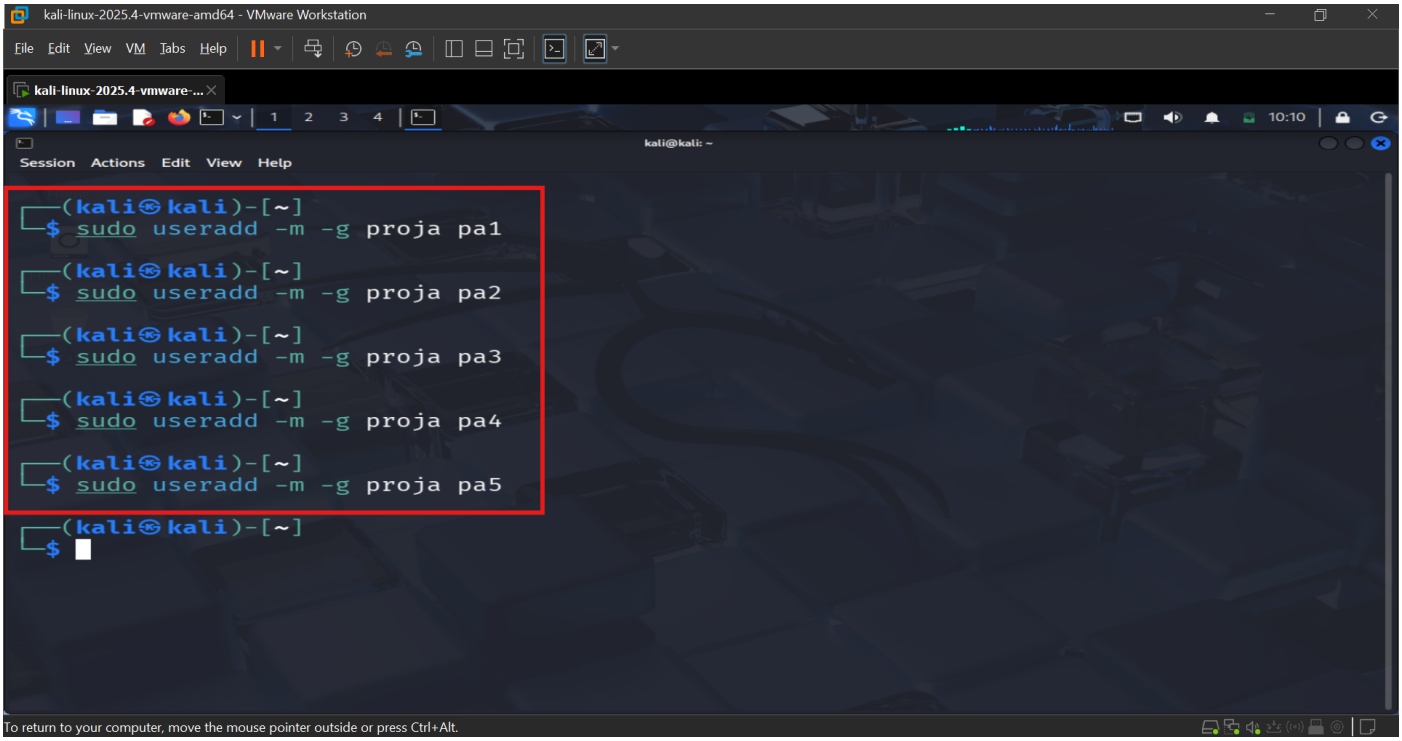


Step 2: Configuring "user accounts" and give permissions

2.1 Create Groups for Project Teams
Command: **sudo groupadd proja**
sudo groupadd projb

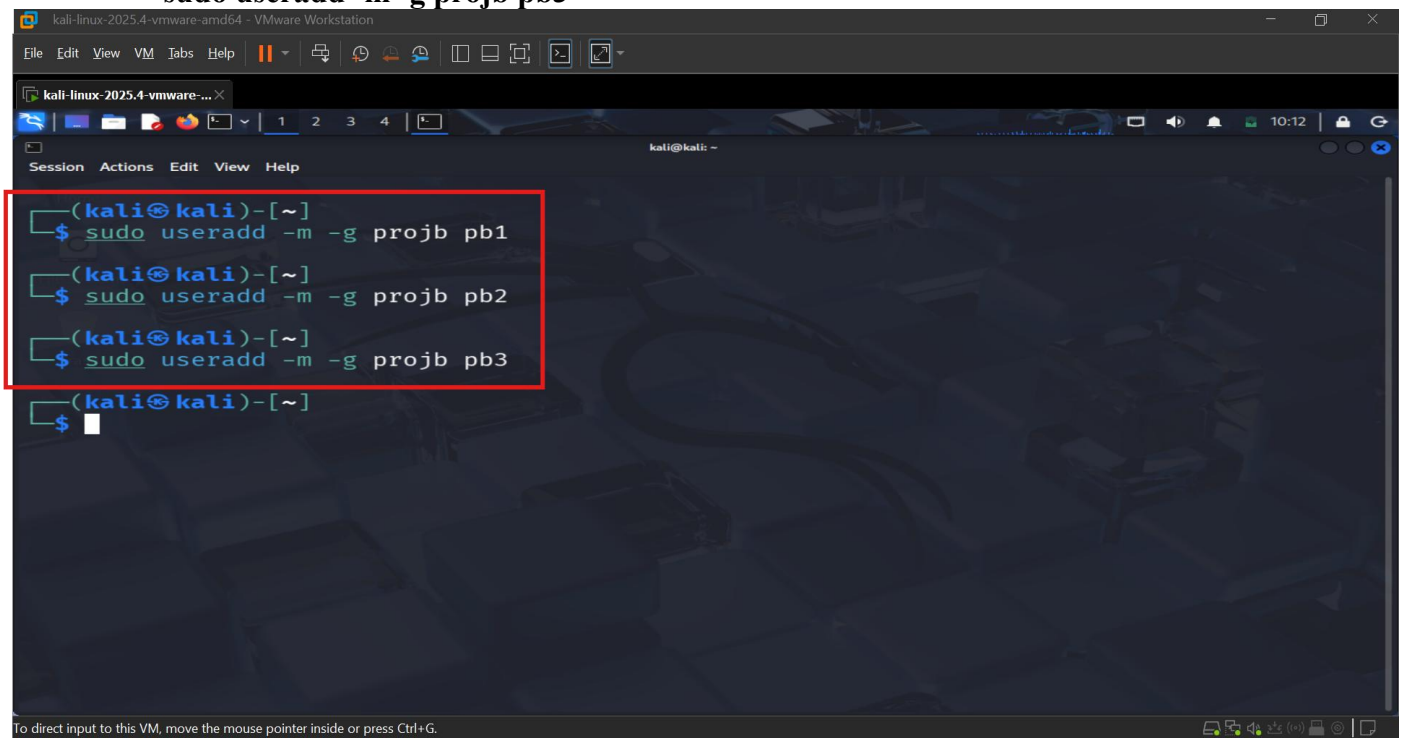


2.2 **Create User Accounts and Assign to Groups**
For Project-A Team (proja users):
Command: sudo useradd -m -g proja pa1
sudo useradd -m -g proja pa2
sudo useradd -m -g proja pa3
sudo useradd -m -g proja pa4
sudo useradd -m -g proja pa5



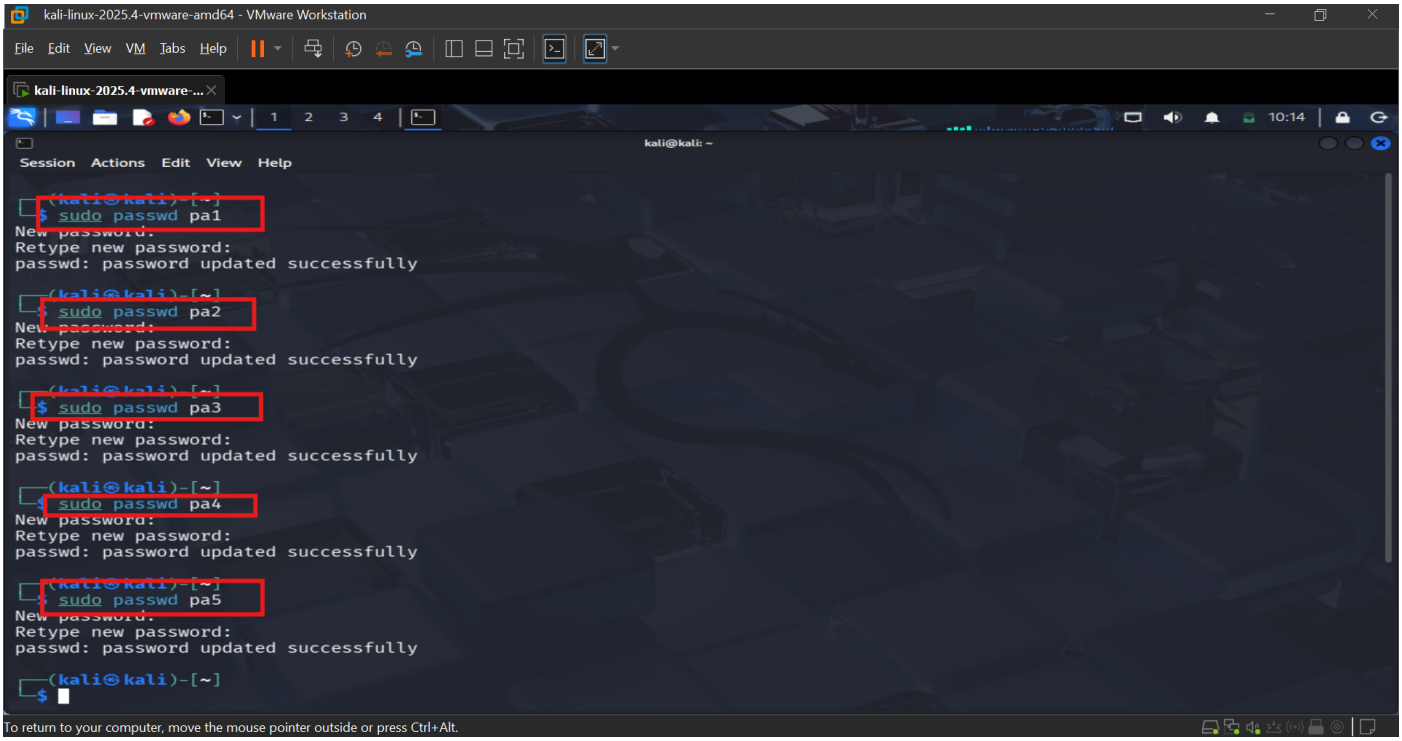
For Project-B Team (projb users):

Command: sudo useradd -m -g projb pb1
sudo useradd -m -g projb pb2
sudo useradd -m -g projb pb3

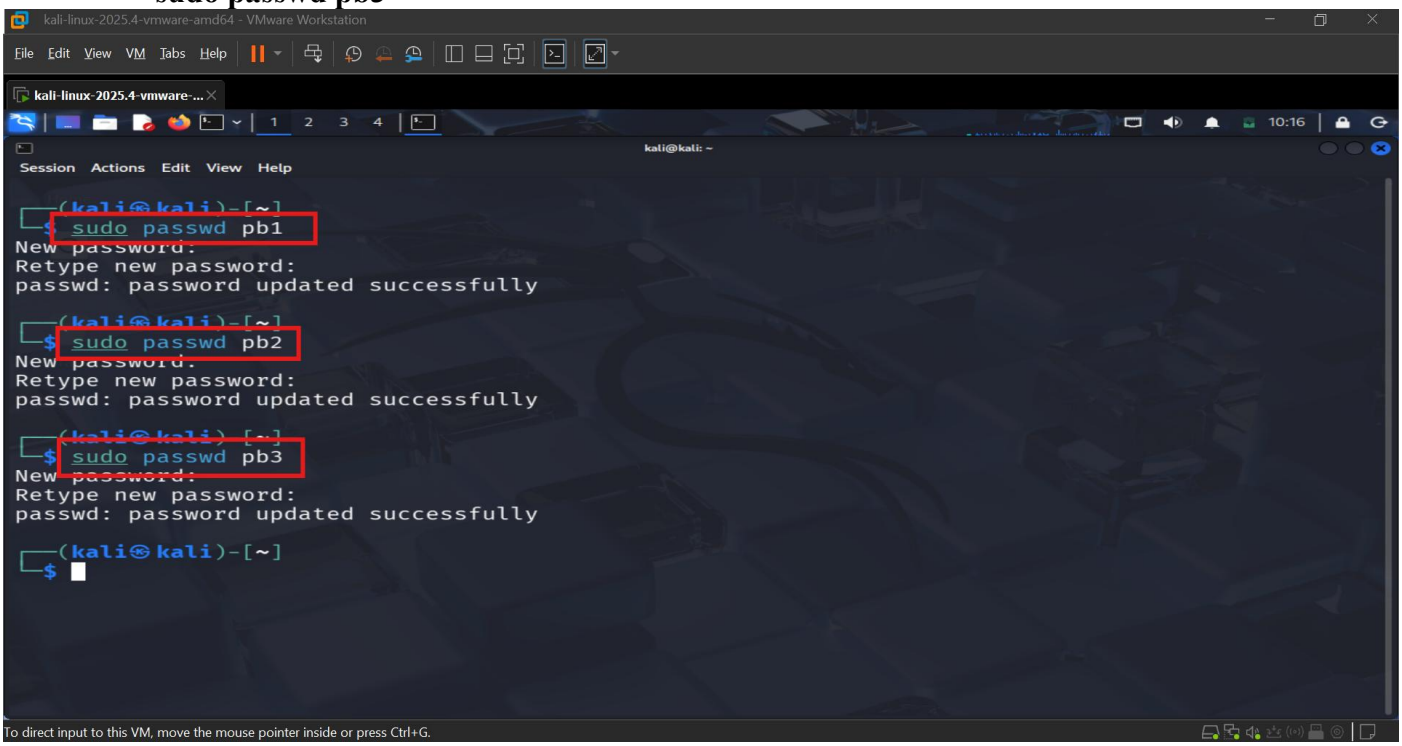


2.3 Set Passwords for Users (pa1,pa2,pa3,pa4,pa5)

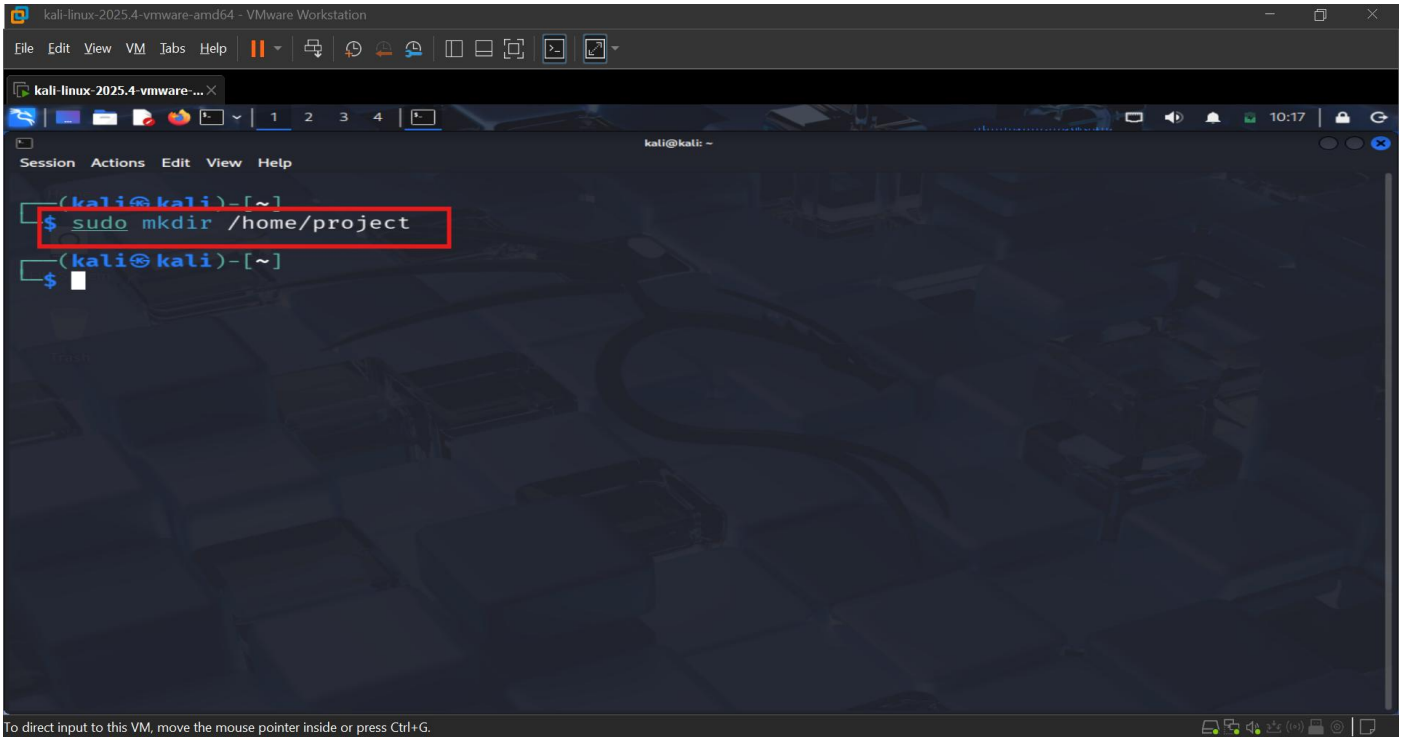
Command: **sudo passwd pa1**
sudo passwd pa2
sudo passwd pa3
sudo passwd pa4
sudo passwd pa5



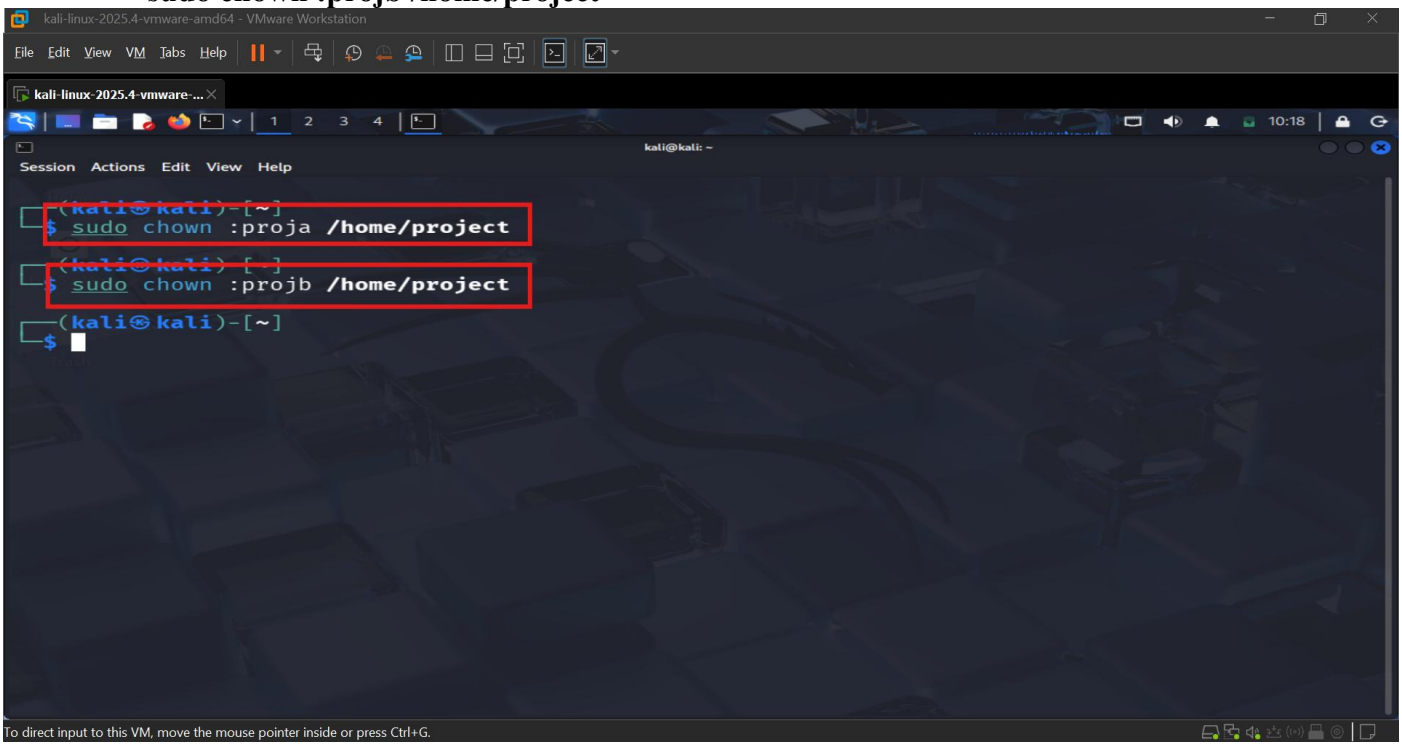
Set Passwords for Users(pb1, pb2, pb3)
Command: **sudo passwd pb1**
sudo passwd pb2
sudo passwd pb3



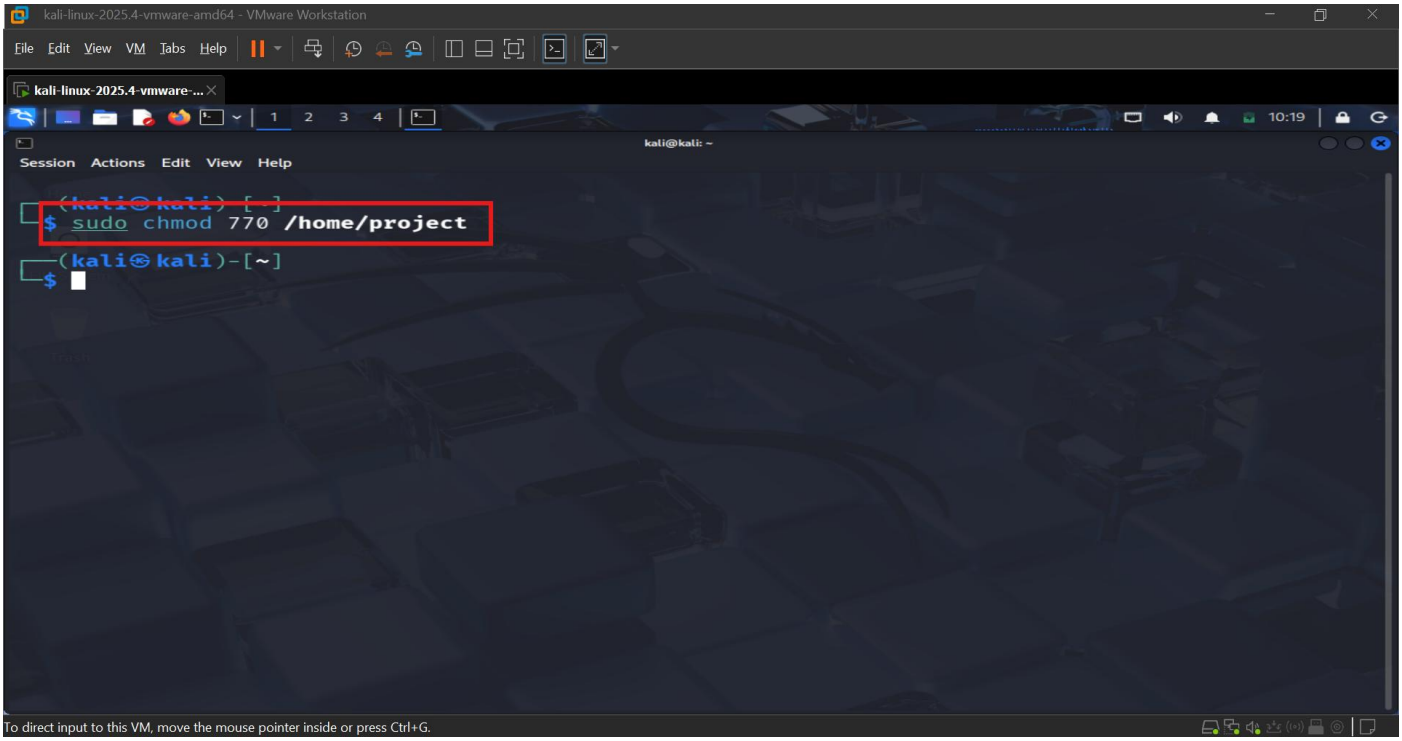
2.4 Create and Secure the Project Directory
Create the Project Directory
Command: **sudo mkdir /home/project**



2.4.1 Assign group ownership to any one of the groups to the directory
Command: **sudo chown :proja /home/project**
sudo chown :projb /home/project

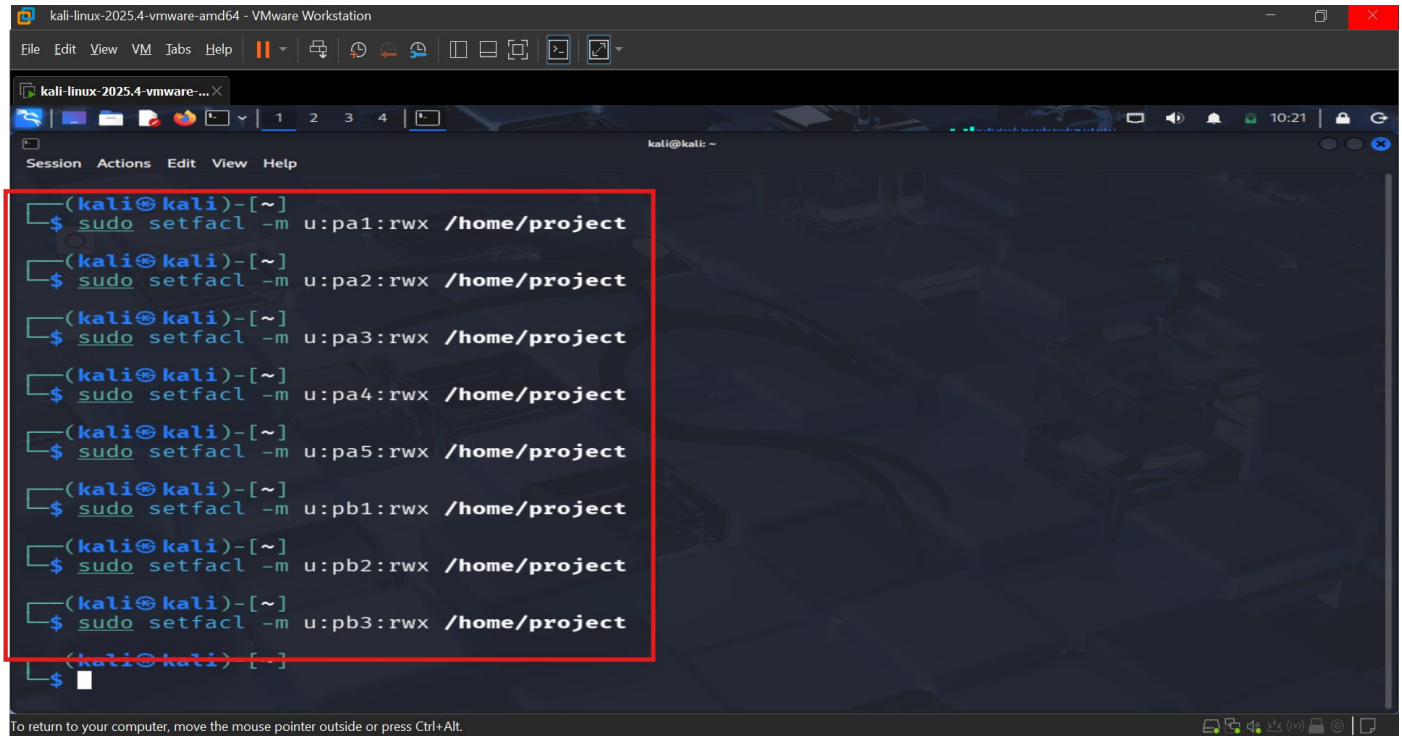


2.4.2 Set Directory Permissions
Command: **sudo chmod 770 /home/project**



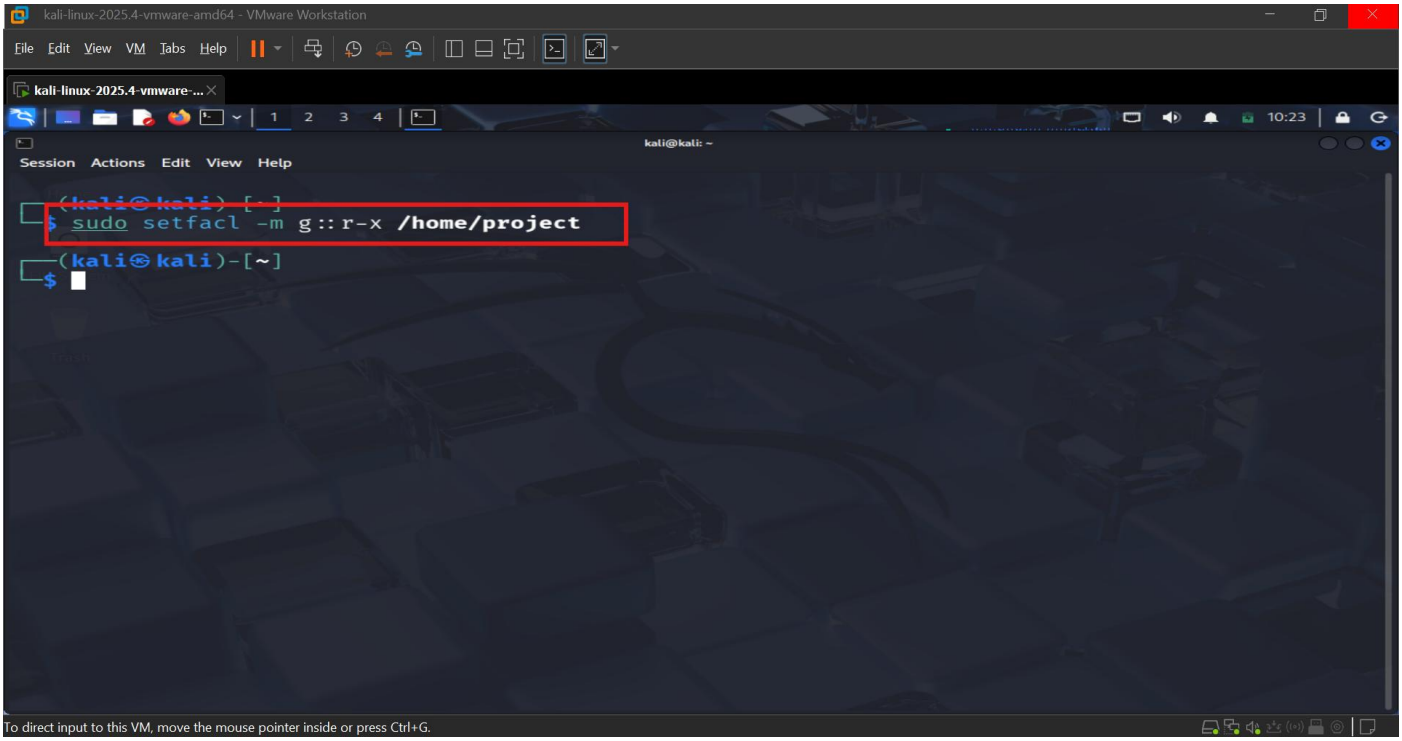
2.5 Configure ACLs to Restrict File Modifications

Command: **sudo setfacl -m u:pa1:rwx /home/project**
sudo setfacl -m u:pa2:rwx /home/project
sudo setfacl -m u:pa3:rwx /home/project
sudo setfacl -m u:pa4:rwx /home/project
sudo setfacl -m u:pa5:rwx /home/project
sudo setfacl -m u:pb1:rwx /home/project
sudo setfacl -m u:pb2:rwx /home/project
sudo setfacl -m u:pb3:rwx /home/project



2.6 Restrict others to delete or modify the file

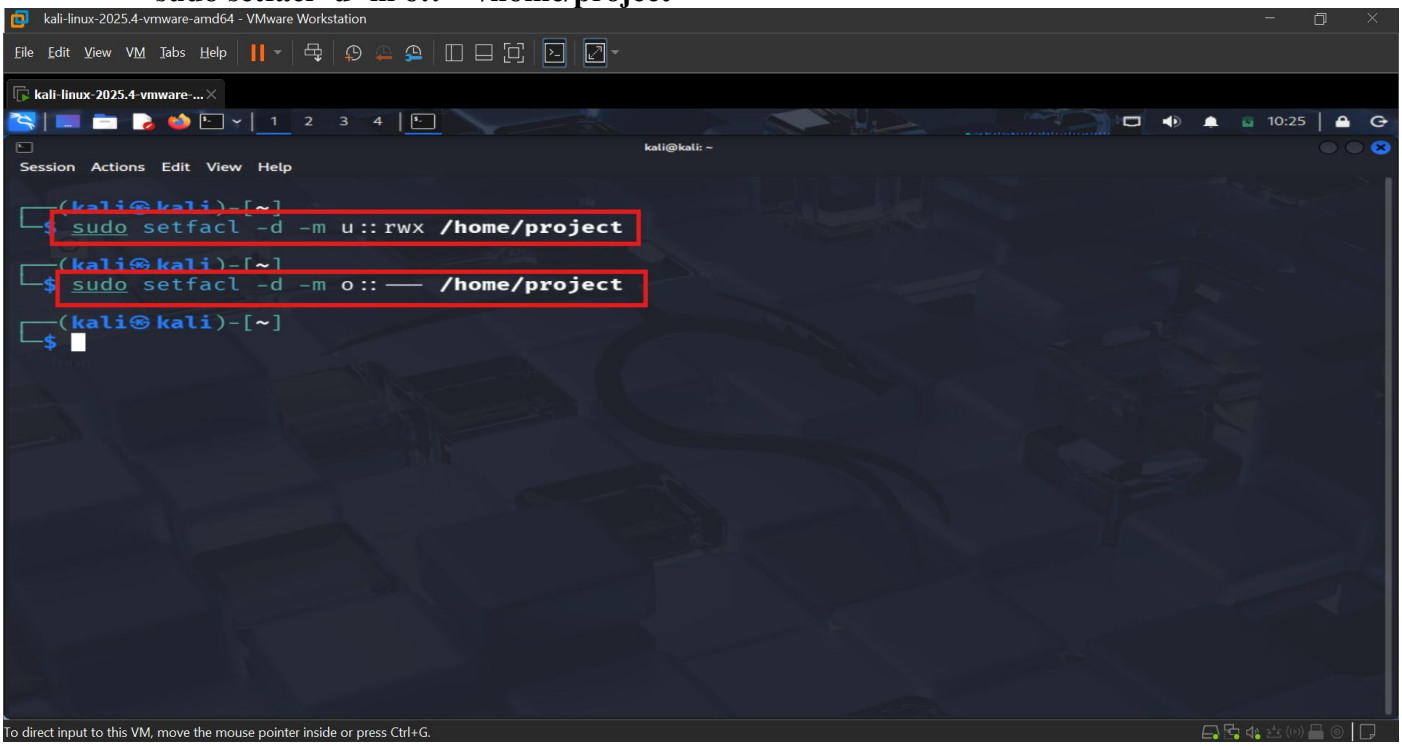
Command: **sudo setfacl -m g::r-x /home/project**



2.7 Apply Default ACLs

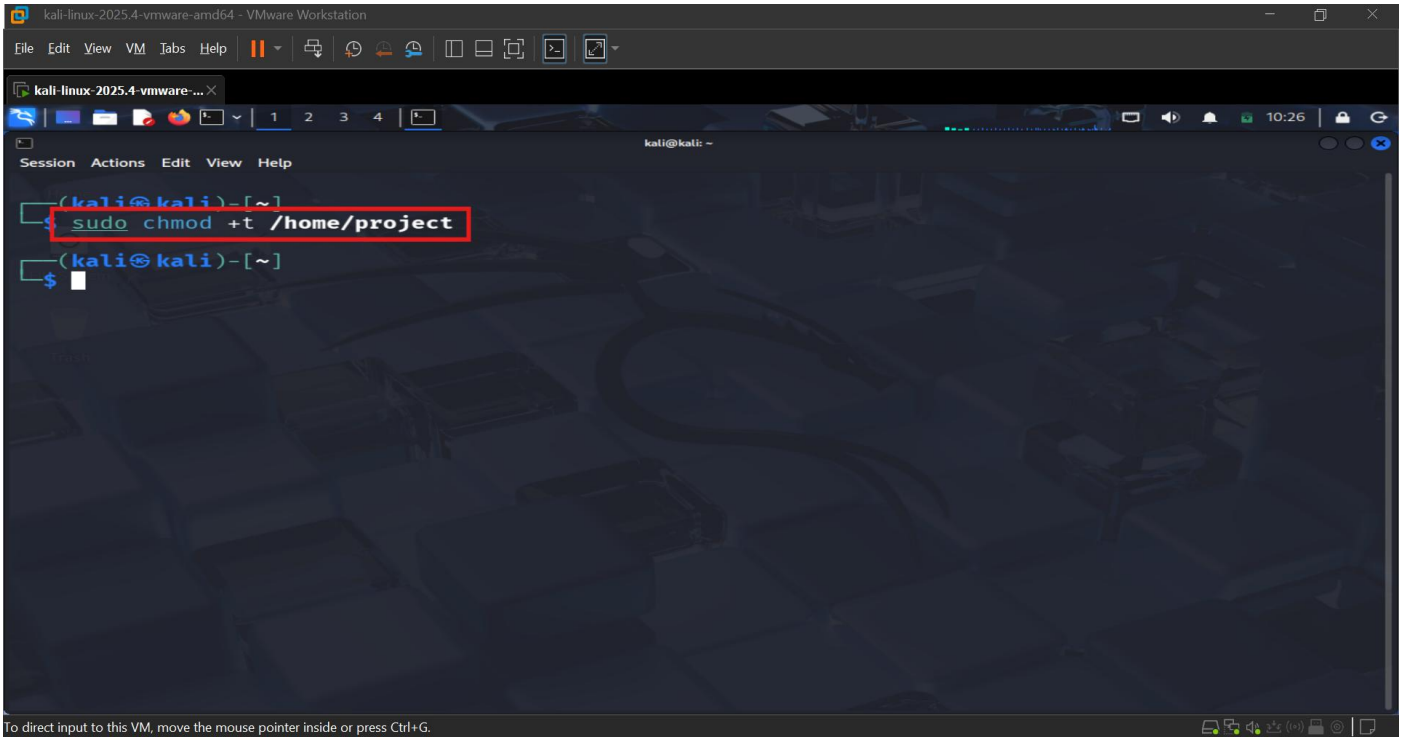
Command: **sudo setfacl -d -m u::rwx /home/project**

sudo setfacl -d -m o::--- /home/project



2.8 Set the Sticky Bit

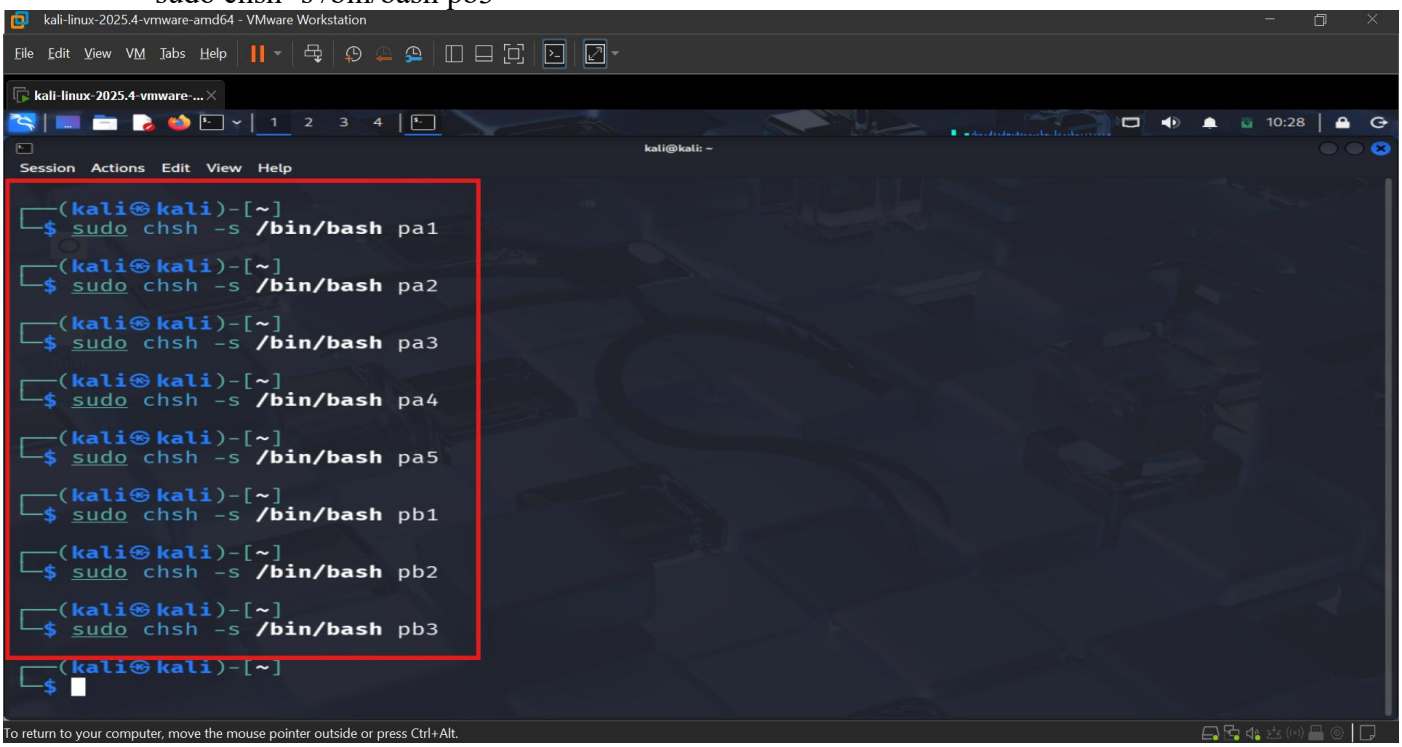
Command: **sudo chmod +t /home/project**



Step 3: Configuring Command History Settings

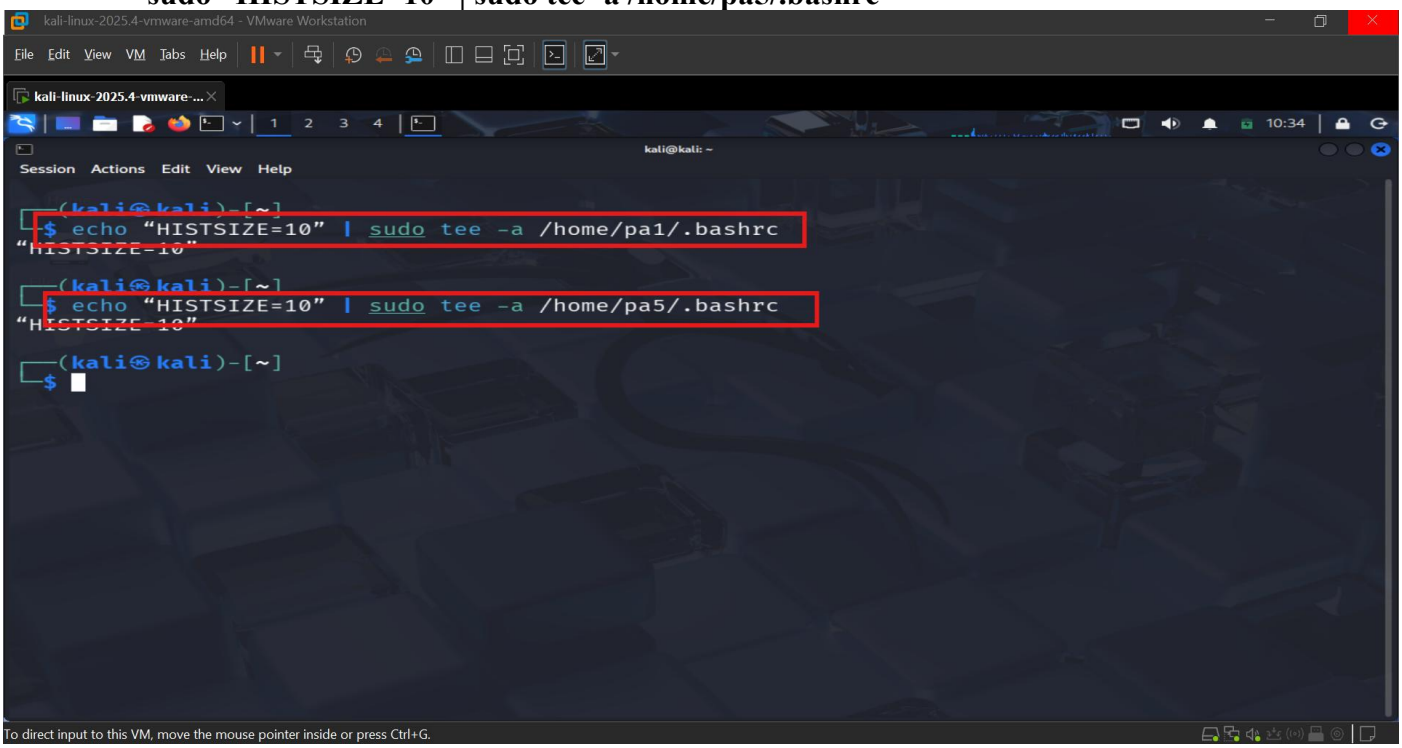
3.1 change the default shell of the user to /bin/bash

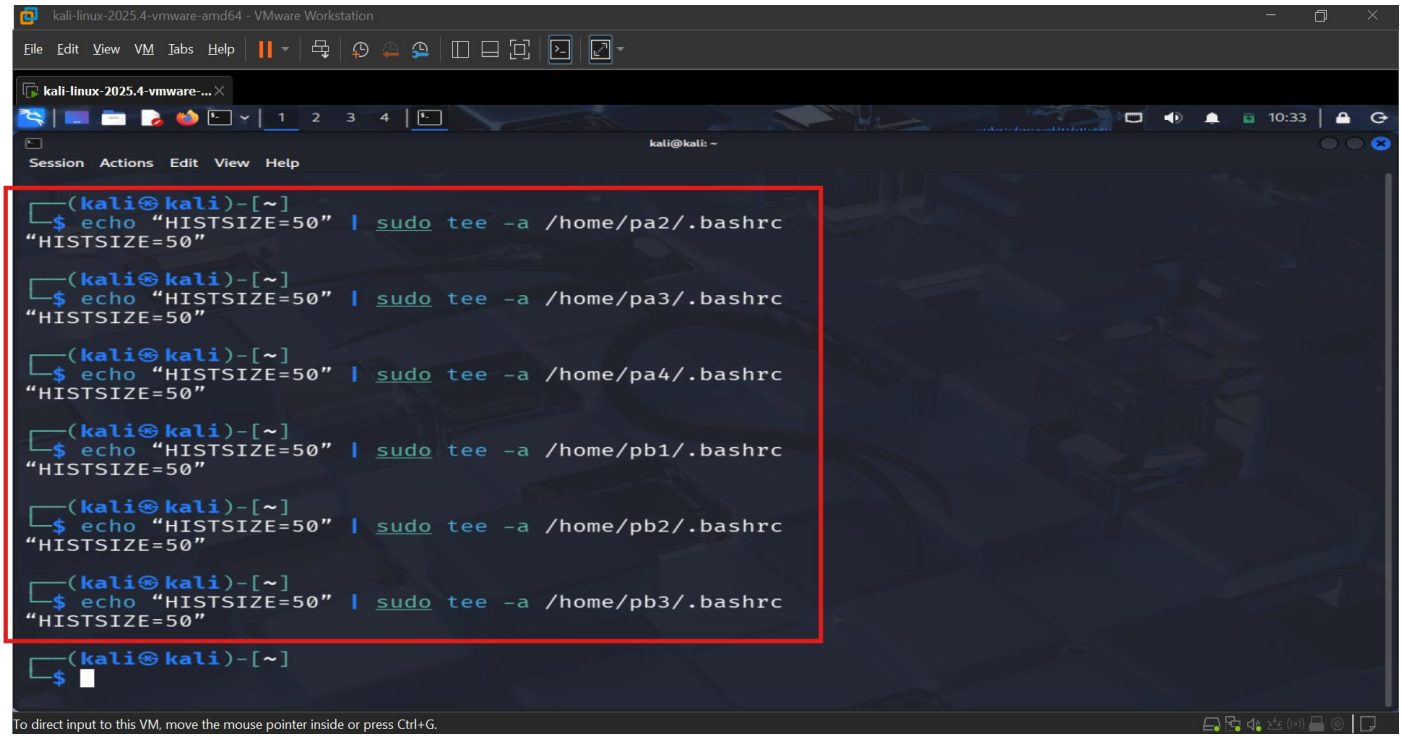
Command: sudo chsh -s /bin/bash pa1
sudo chsh -s /bin/bash pa2
sudo chsh -s /bin/bash pa3
sudo chsh -s /bin/bash pa4
sudo chsh -s /bin/bash pa5
sudo chsh -s /bin/bash pb1
sudo chsh -s /bin/bash pb2
sudo chsh -s /bin/bash pb3



3.2 Set Command History Limits for Users(for pa1 and pa5)

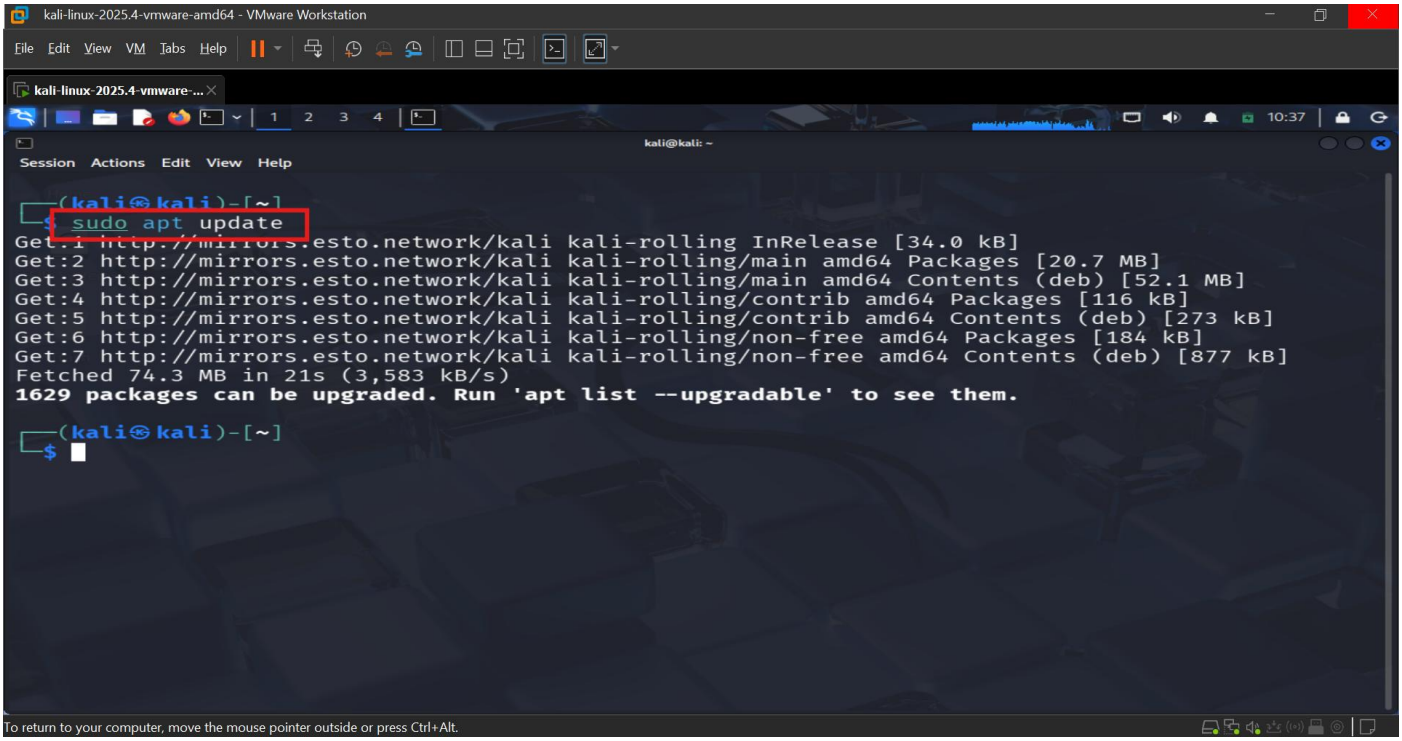
Command: **sudo echo “HISTSIZE=10” | sudo tee -a /home/pa1/.bashrc**
sudo “HISTSIZE=10” | sudo tee -a /home/pa5/.bashrc





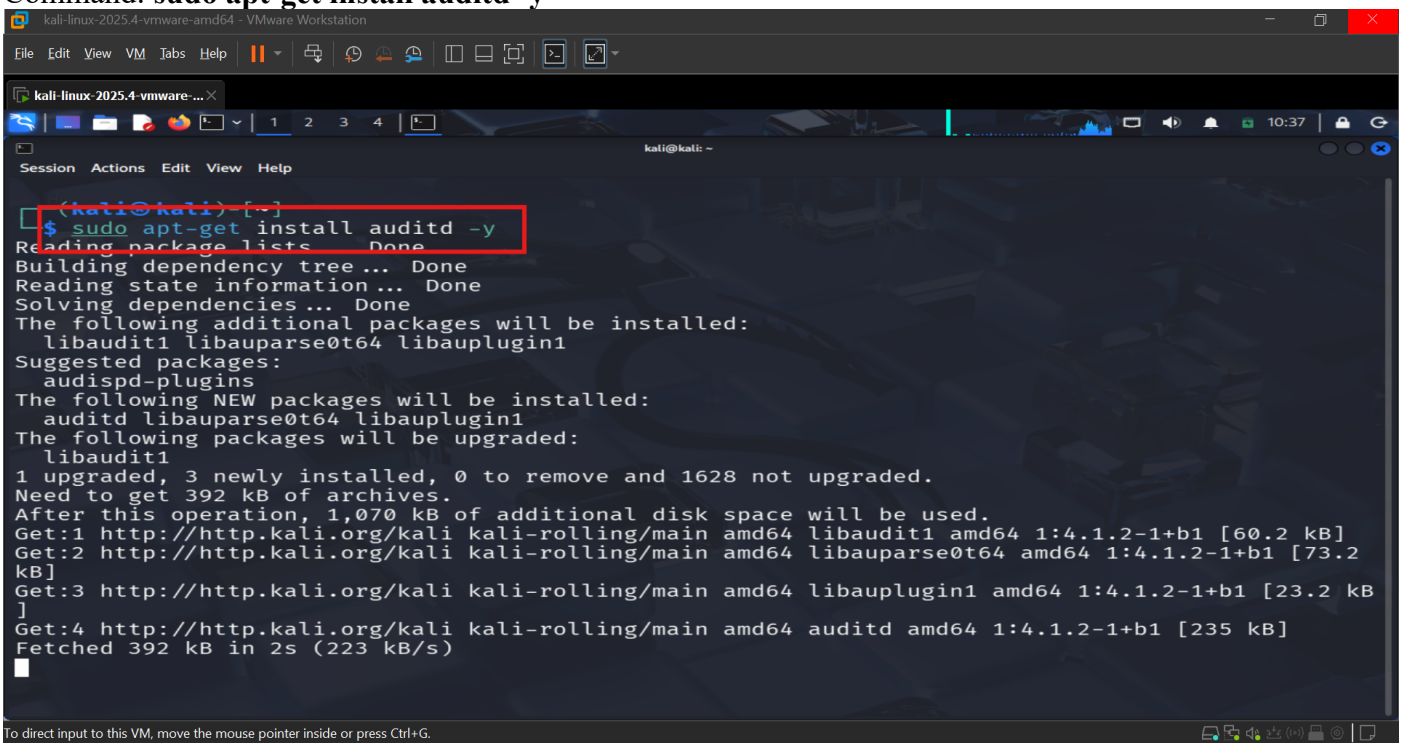
Step 4: Logging Security Violations

4.1 Update the os
Command: **sudo apt update**



4.2 Install and Configure Auditd

Command: **sudo apt-get install auditd -y**



4.3 Create Audit Rules to Log Access to the Project Directory

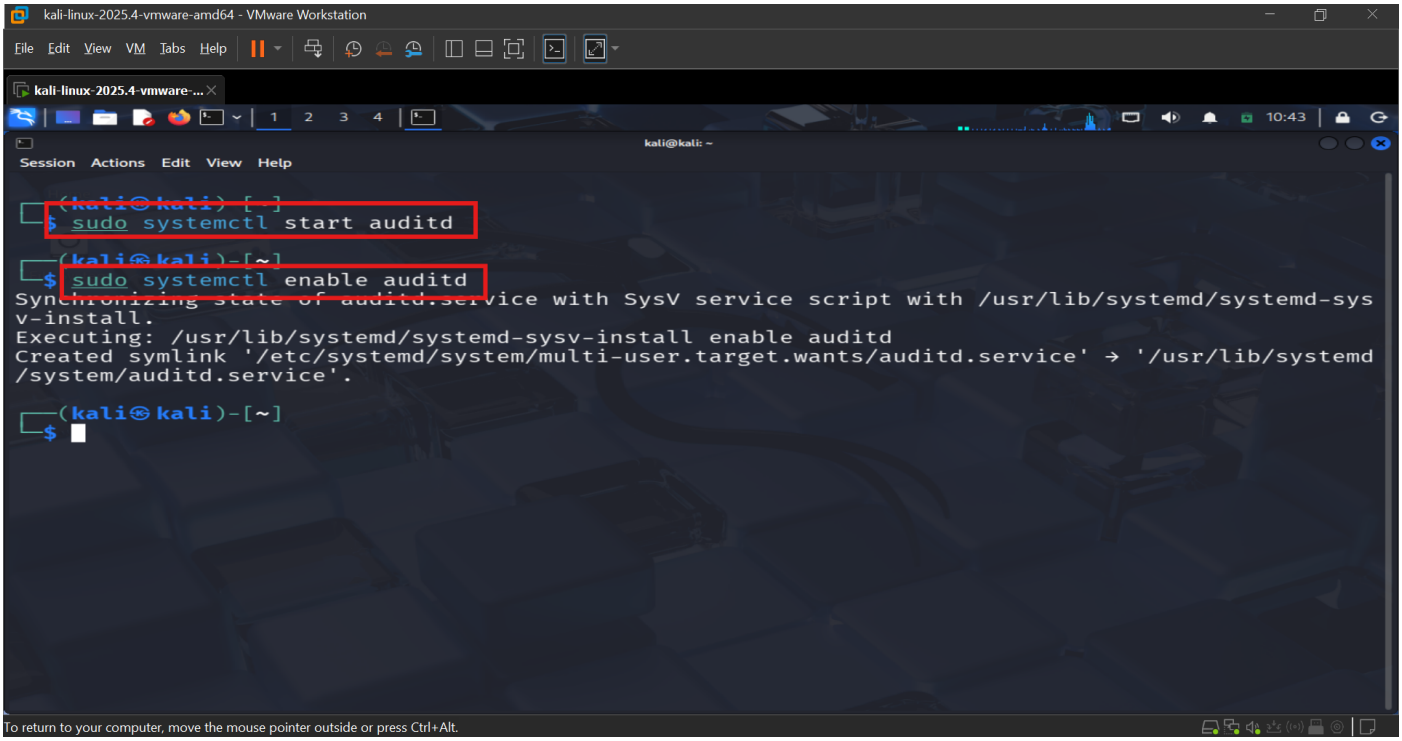
Command: **sudo auditctl -w /home/project -p rwxa -k project_access**

```
kali-linux-2025.4-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali@kali: ~
Session Actions Edit View Help
o.0.usr-is-merged by libauparse0t64'
Adding 'diversion of /lib/x86_64-linux-gnu/libauparse.so.0.0.0 to /lib/x86_64-linux-gnu/libaupar
se.so.0.0.0.usr-is-merged by libauparse0t64'
Unpacking libauparse0t64:amd64 (1:4.1.2-1+b1) ...
Selecting previously unselected package libauplugin1:amd64.
Preparing to unpack .../libauplugin1_1%3a4.1.2-1+b1_amd64.deb ...
Unpacking libauplugin1:amd64 (1:4.1.2-1+b1) ...
Selecting previously unselected package auditd.
Preparing to unpack .../auditd_1%3a4.1.2-1+b1_amd64.deb ...
Unpacking auditd (1:4.1.2-1+b1) ...
Setting up libauparse0t64:amd64 (1:4.1.2-1+b1) ...
Setting up libauplugin1:amd64 (1:4.1.2-1+b1) ...
Setting up auditd (1:4.1.2-1+b1) ...
update-rc.d: We have no instructions for the auditd init script.
update-rc.d: It looks like a non-network service, we enable it.
audit-rules.service is a disabled or a static unit, not starting it.
auditd.service is a disabled or a static unit, not starting it.
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for kali-menu (2025.4.3) ...
Processing triggers for libc-bin (2.41-12) ...

┌──(kali㉿kali)-[~]
└─$ sudo auditctl -w /home/project -p rwxa -k project_access
Old style watch rules are slower

┌──(kali㉿kali)-[~]
└─$
To return to your computer, move the mouse pointer outside or press Ctrl+Alt.
```

4.4 Install the Audit Daemon (auditd) dependencies.
Command: **sudo apt install auditd audispd-plugins -y**

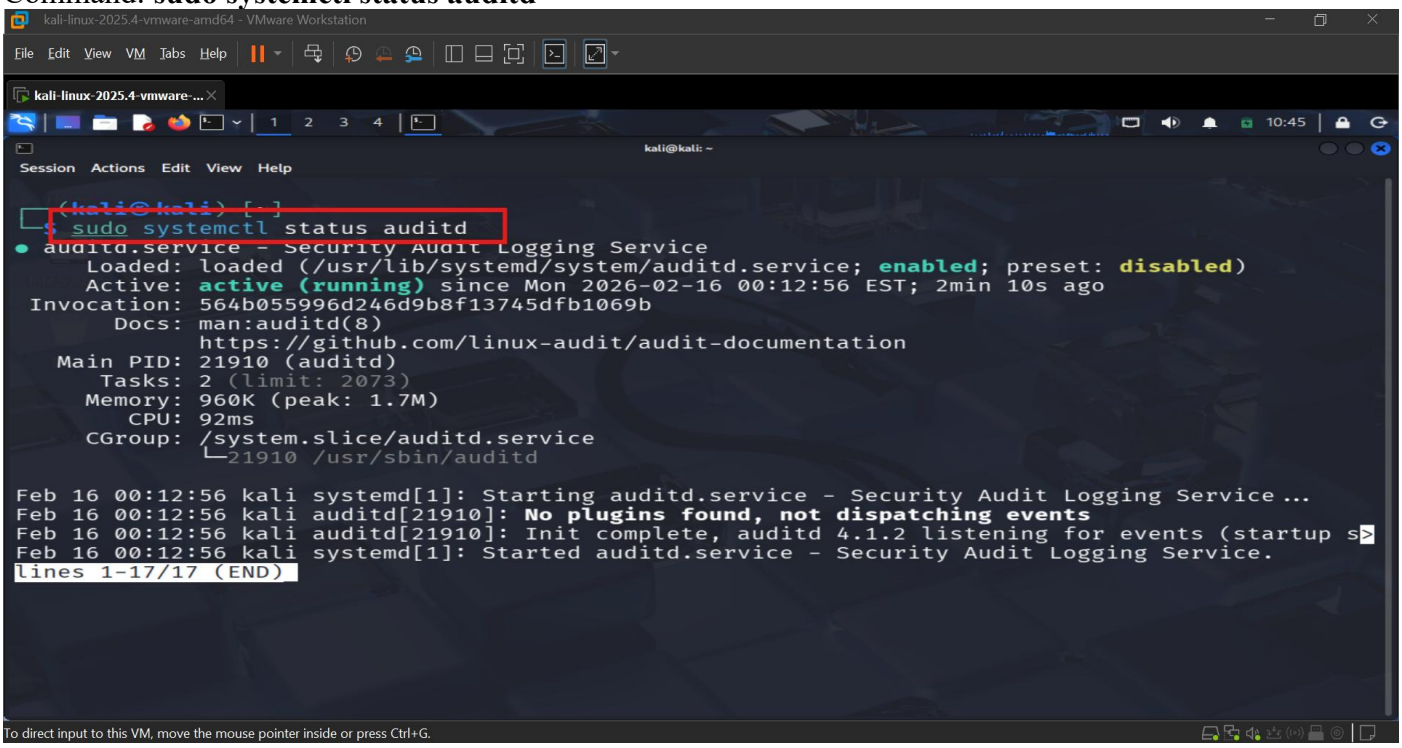
```
kali-linux-2025.4-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali@kali: ~
Session Actions Edit View Help

┌──(kali㉿kali)-[~]
└─$ sudo apt install auditd audispd-plugins -y
auditd is already the newest version (1:4.1.2-1+b1).
Installing:
  audispd-plugins

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1628
  Download size: 51.8 kB
  Space needed: 145 kB / 63.0 GB available

Get:1 http://http.kali.org/kali kali-rolling/main amd64 audispd-plugins amd64 1:4.1.2-1+b1 [51.8
 kB]
Fetched 51.8 kB in 1s (41.7 kB/s)
Selecting previously unselected package audispd-plugins.
(Reading database ... 422418 files and directories currently installed.)
Preparing to unpack .../audispd-plugins_1%3a4.1.2-1+b1_amd64.deb ...
Unpacking audispd-plugins (1:4.1.2-1+b1) ...
Setting up audispd-plugins (1:4.1.2-1+b1) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for kali-menu (2025.4.3) ...

┌──(kali㉿kali)-[~]
└─$
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
```

4.5 Start and Enable the auditd Service
Command: **sudo systemctl start auditd**
sudo systemctl enable auditd



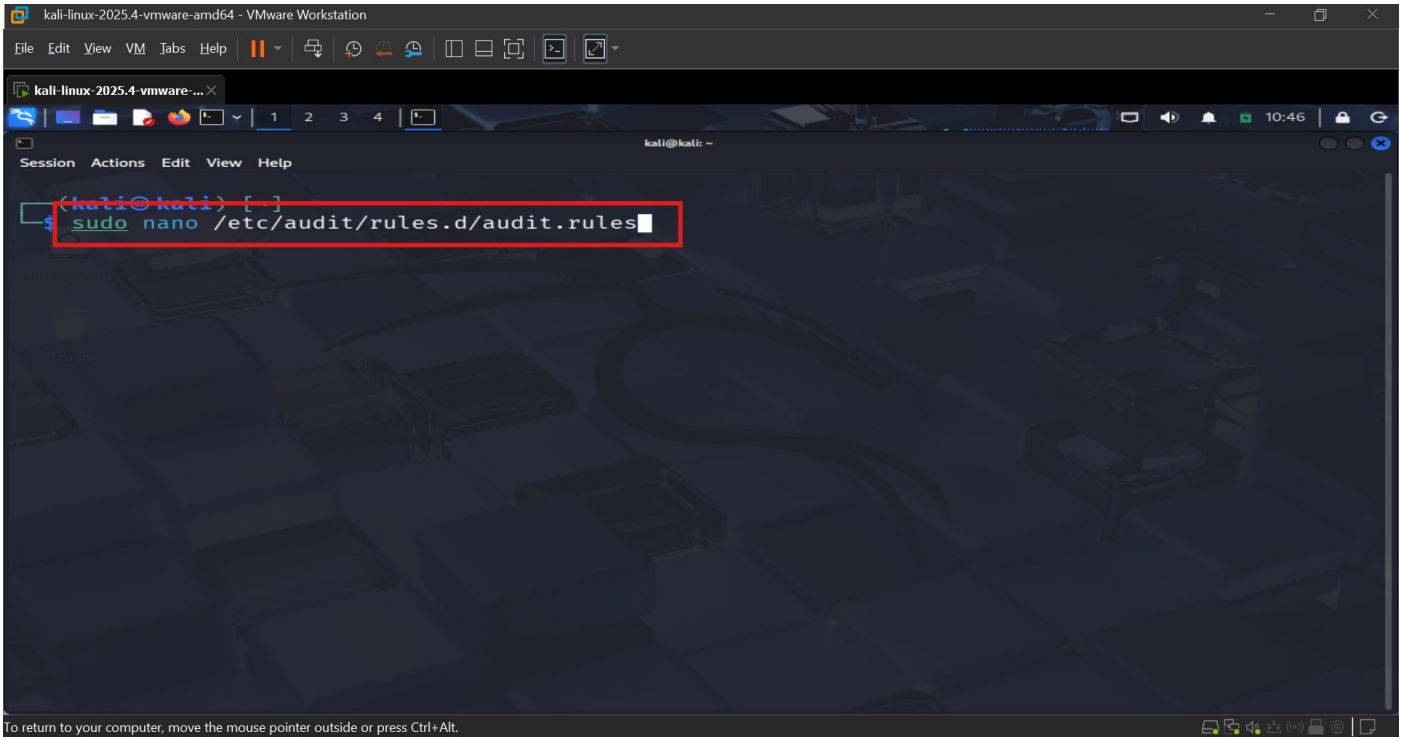
4.6 Verify auditd Service Status

Command: **sudo systemctl status auditd**

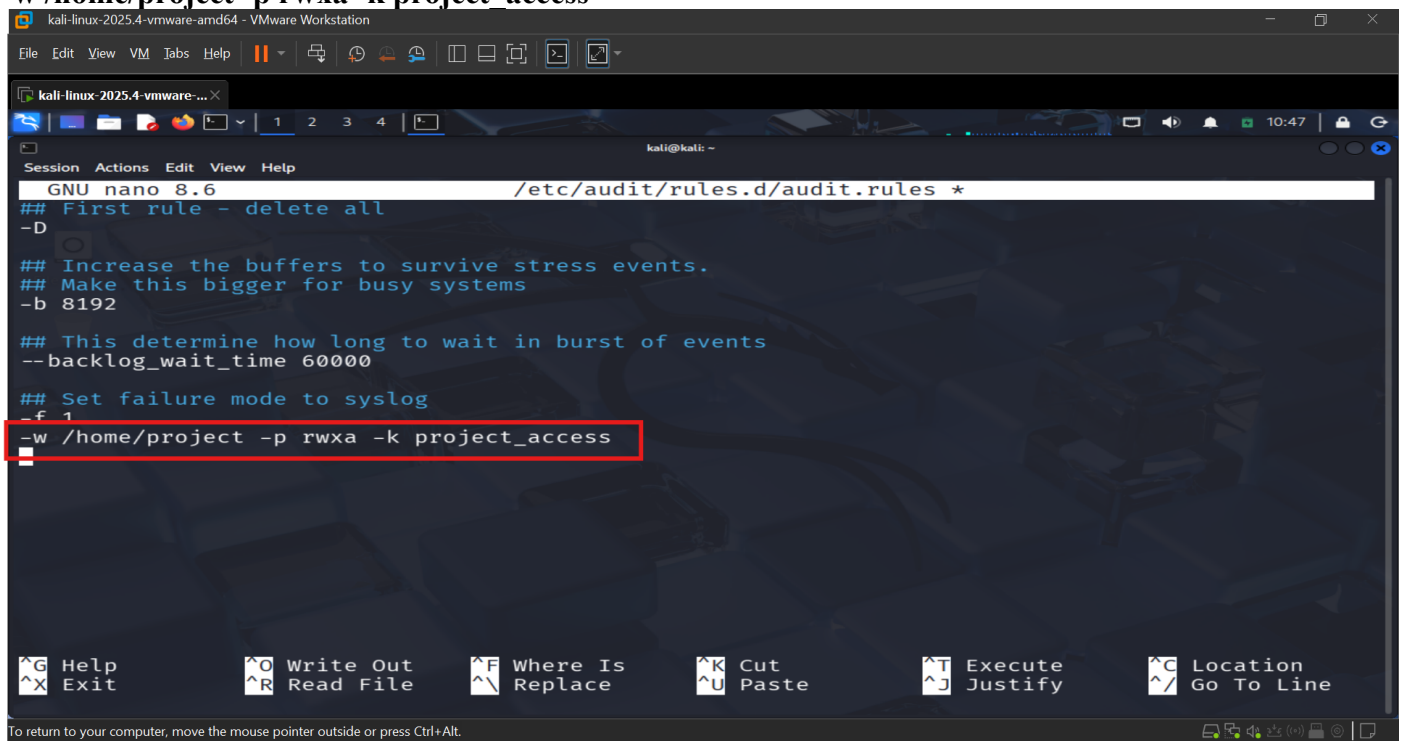


4.7 Create and Verify Audit Rules

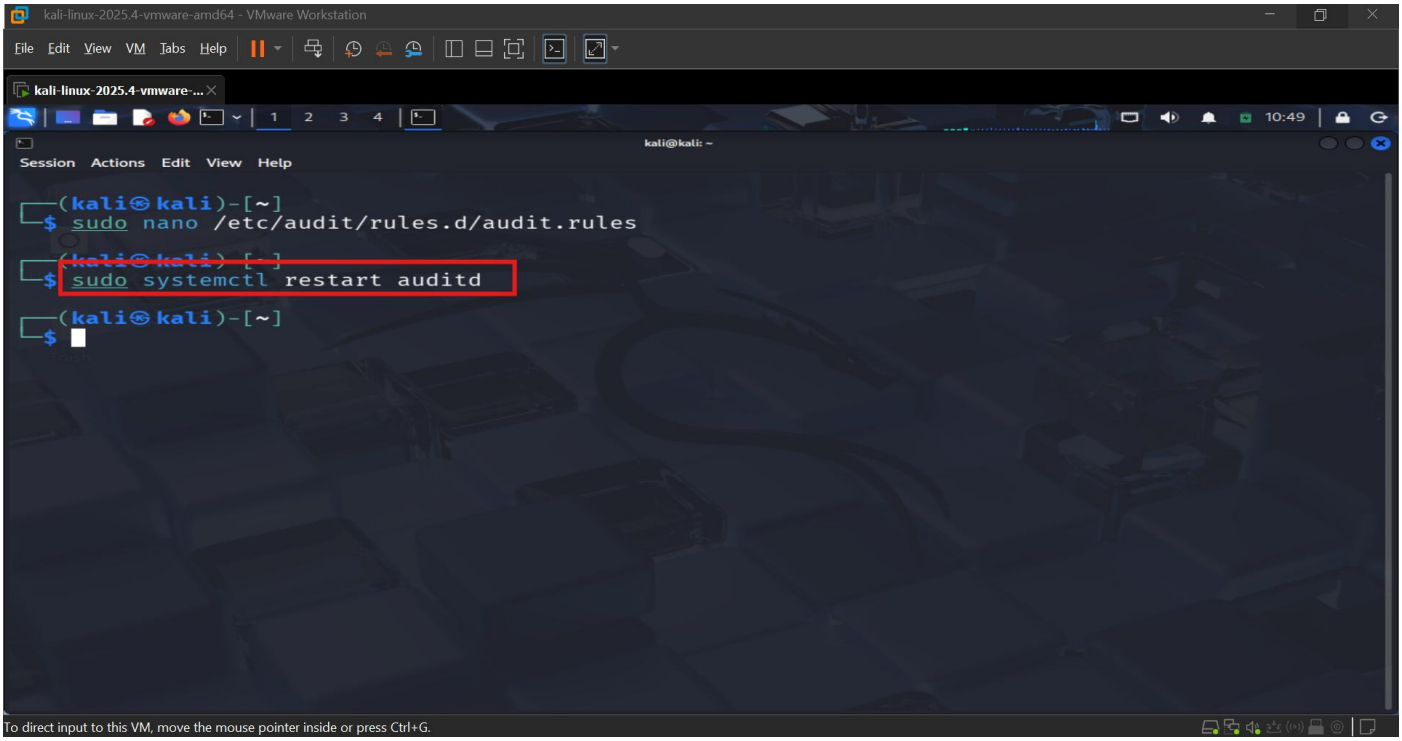
Command: **sudo nano /etc/audit/rules.d/audit.rules**



Add the following line at the end of the file
-w /home/project -p rwxa -k project_access

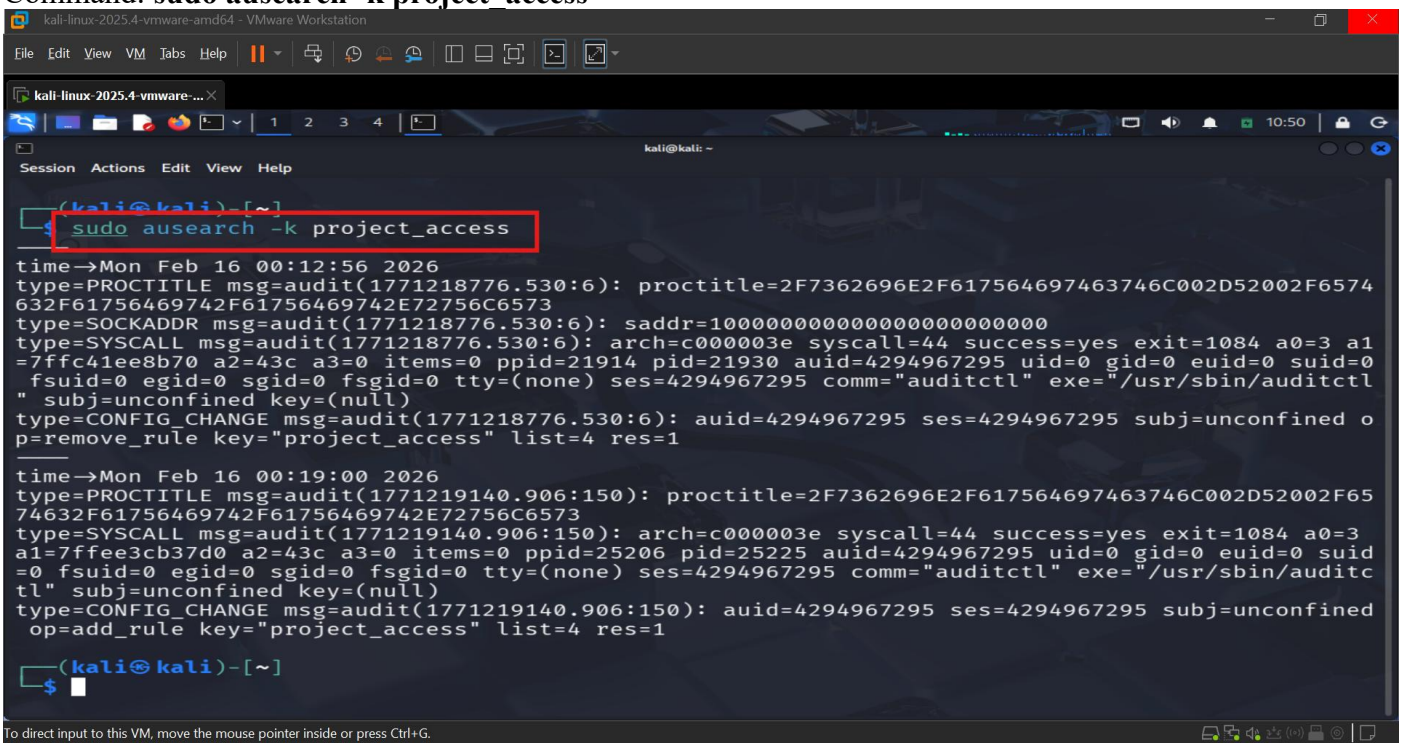


4.8 Restart the auditd service to apply the changes
Command: **sudo systemctl restart auditd**



4.9 Test the Configuration

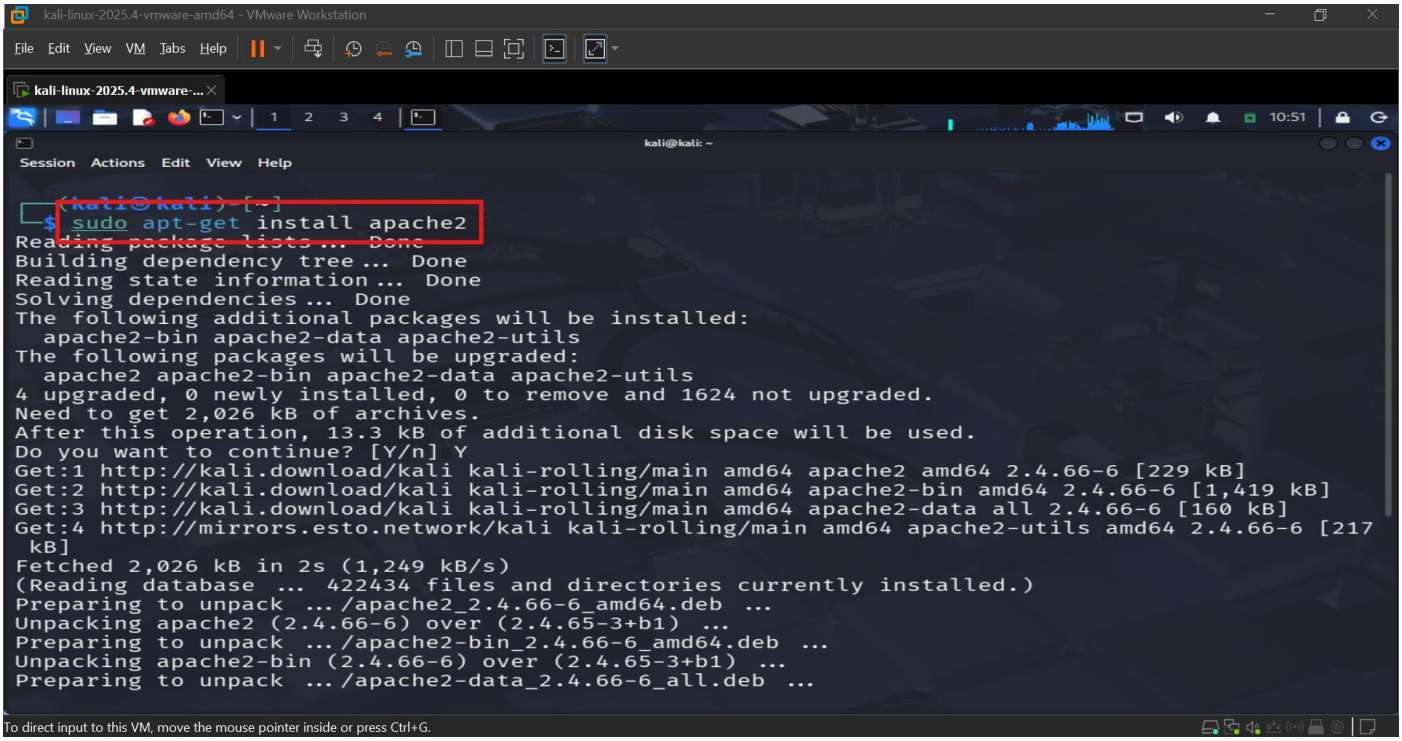
Command: **sudo ausearch -k project_access**



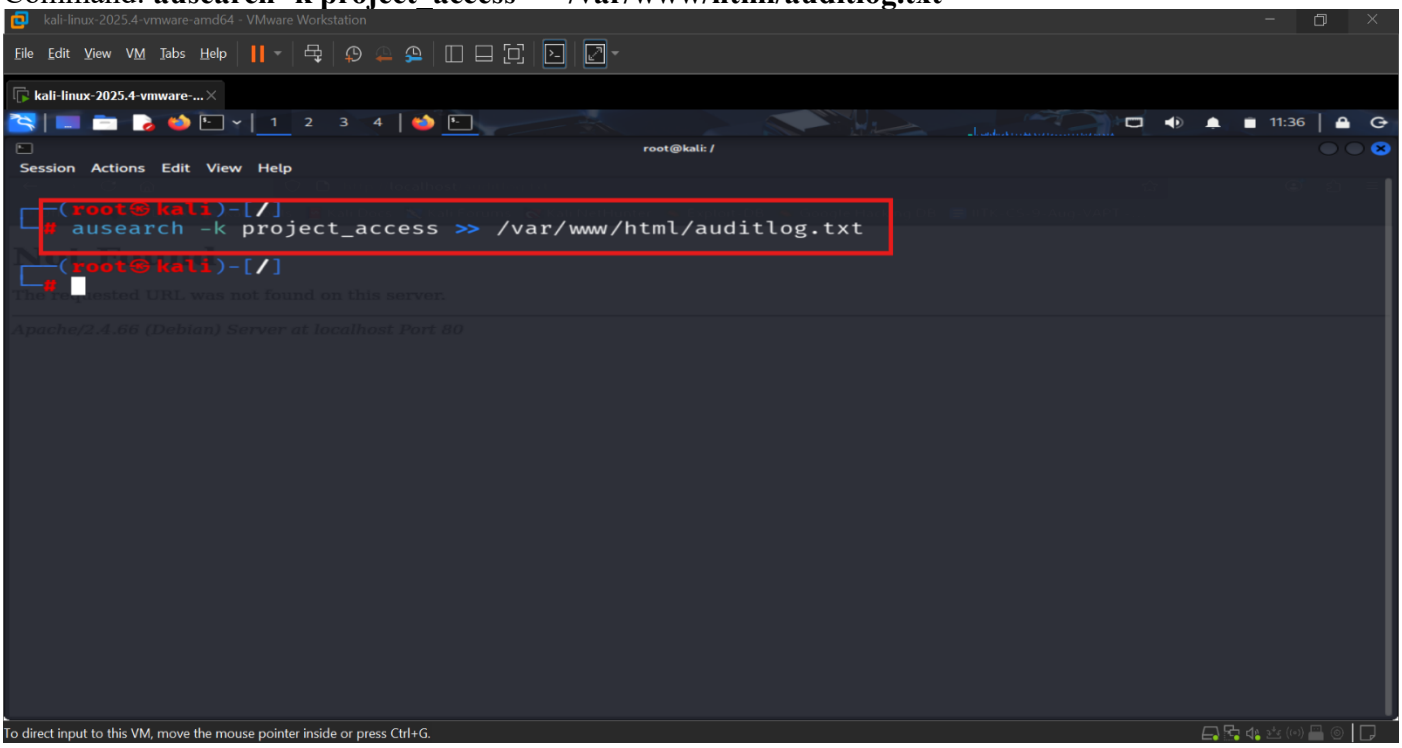
Step 5: Developing a Web-Based Report System

5.1 Installing Apache2

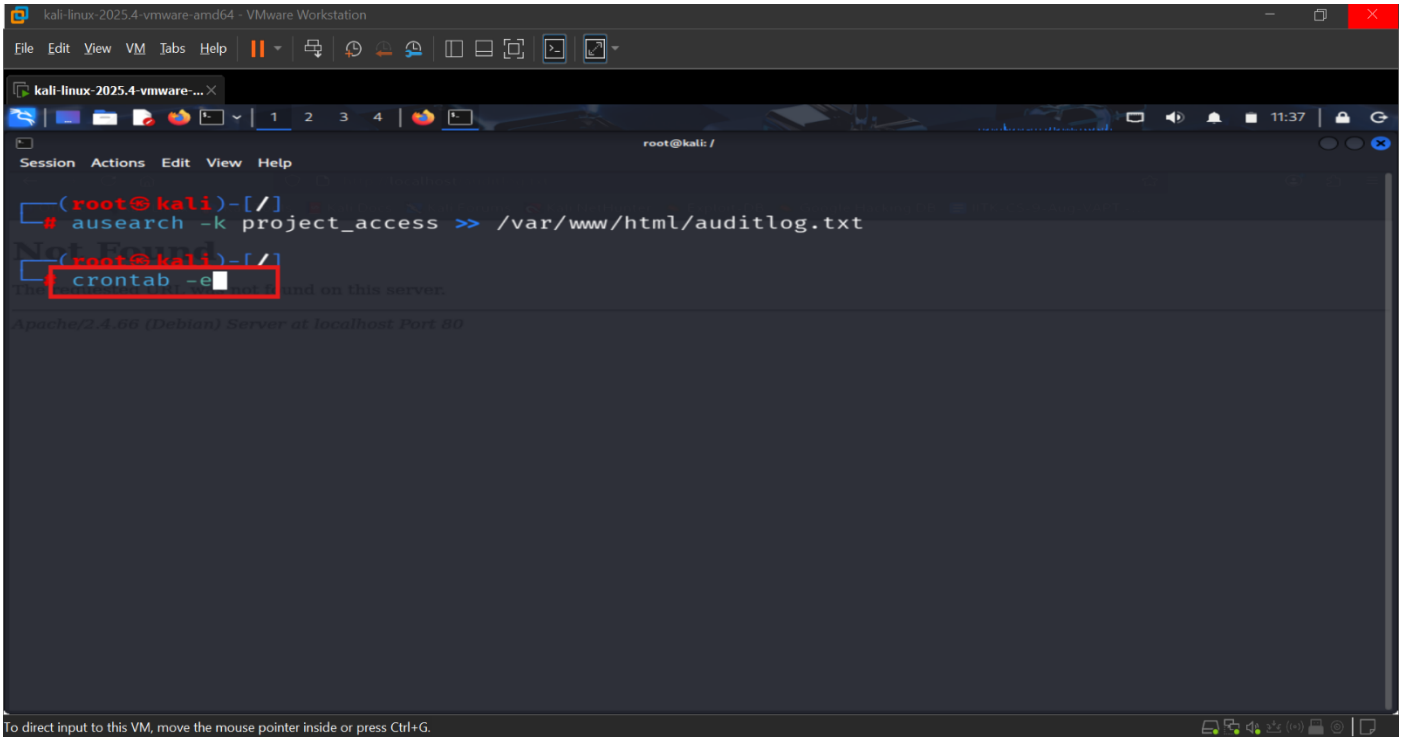
Command: **sudo apt-get install apache2**



5.2 Run ausearch Command and transfer the logs to a
Command: **ausearch -k project_access >> /var/www/html/auditlog.txt**

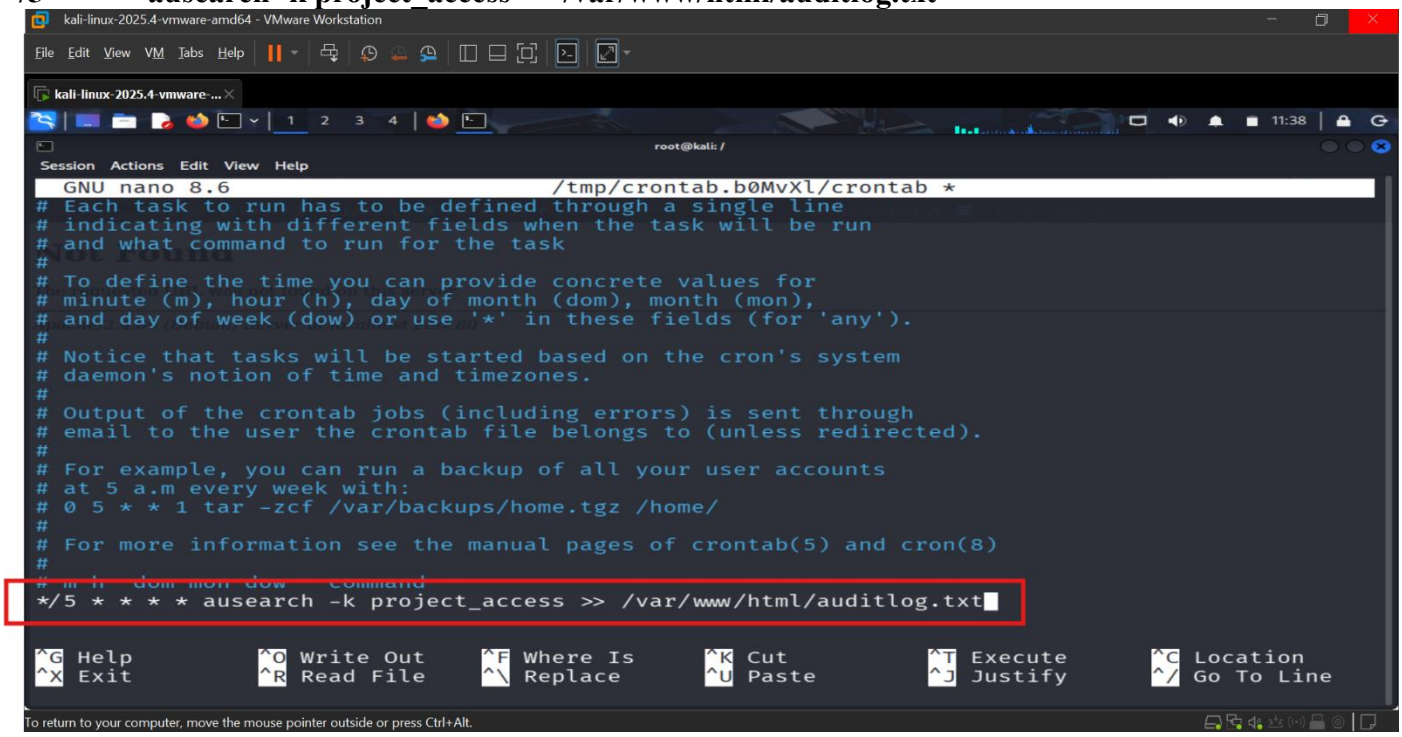


5.3 Using Crontab for Automation
Command: **crontab -e**



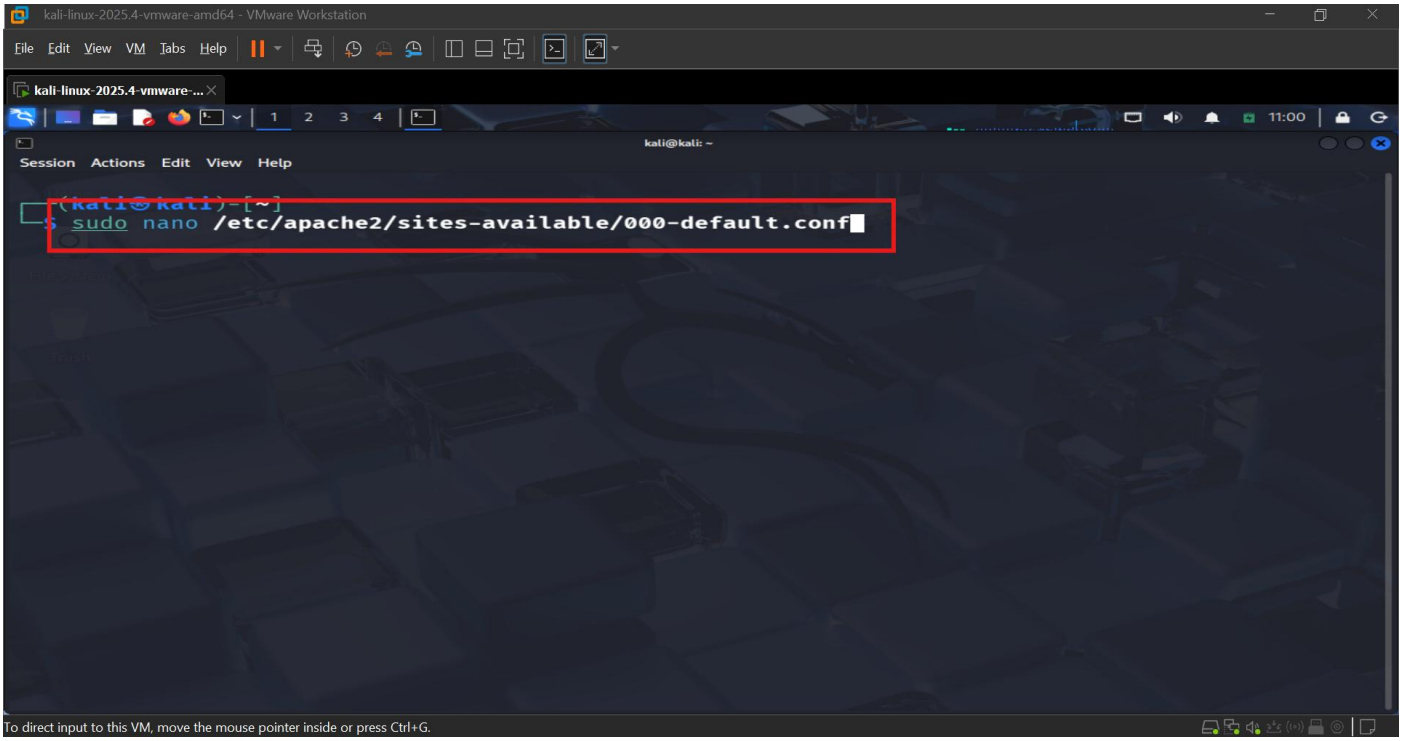
Within the crontab file, add this line

***/5 * * * * ausearch -k project_access >> /var/www/html/auditlog.txt**

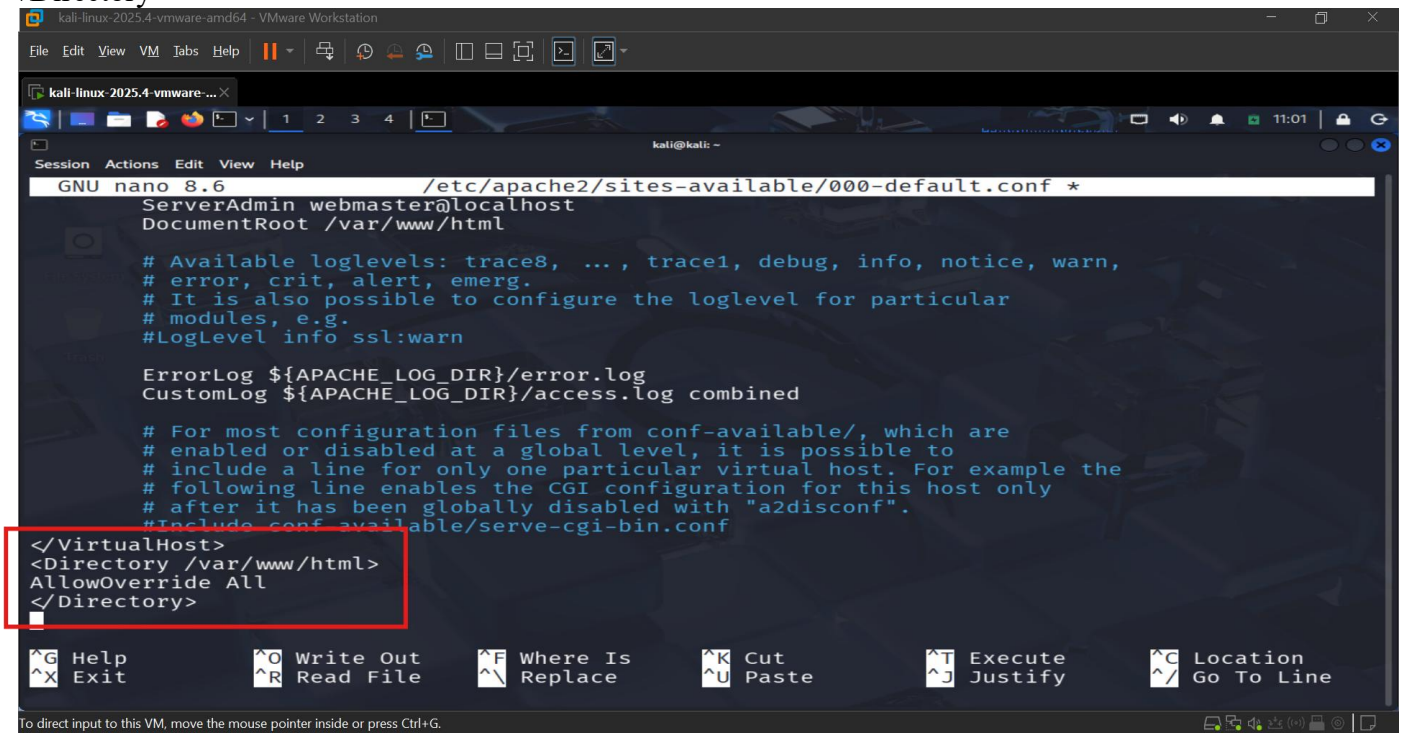


5.4 Edit the Apache configuration file

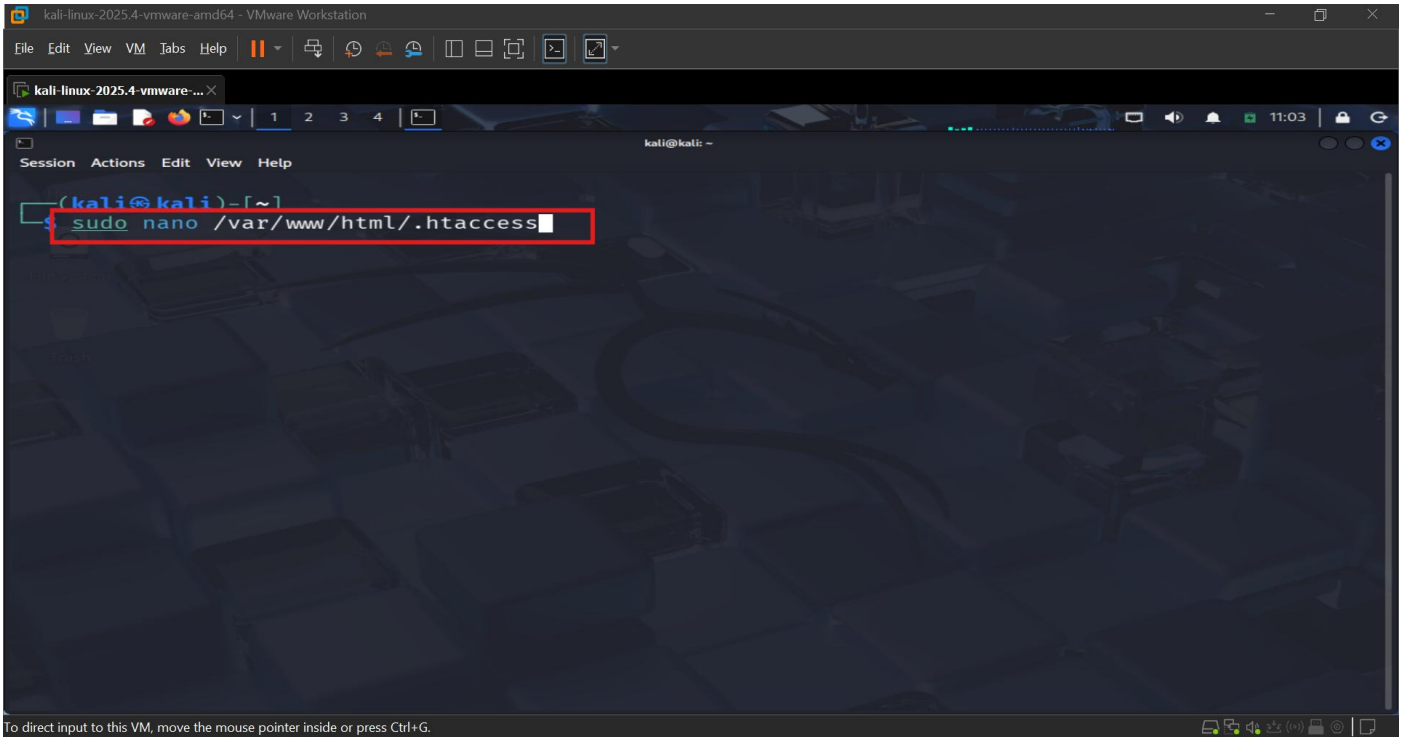
Command: **sudo nano /etc/apache2/sites-available/000-default.conf**



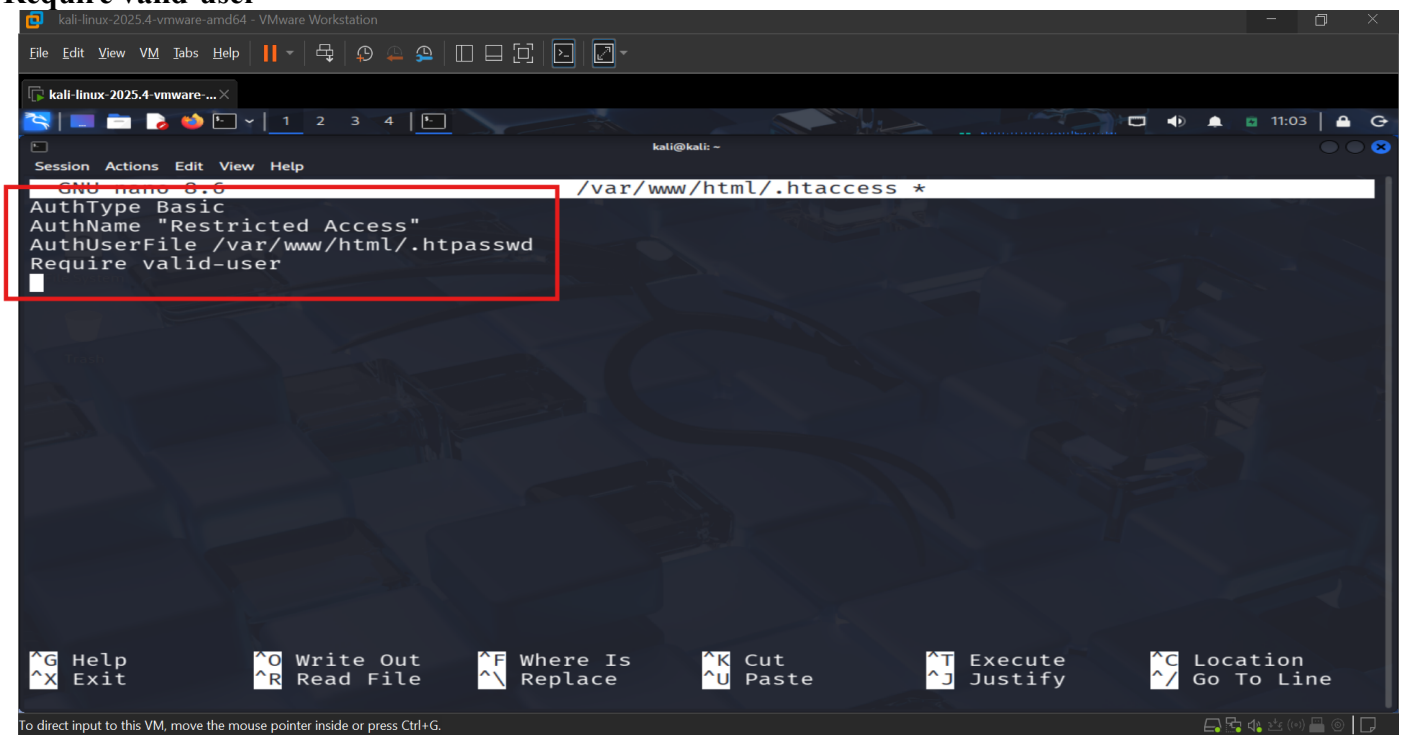
Add or modify the following lines:
</VirtualHost>
<Directory /var/www/html>
AllowOverride All
</Directory>



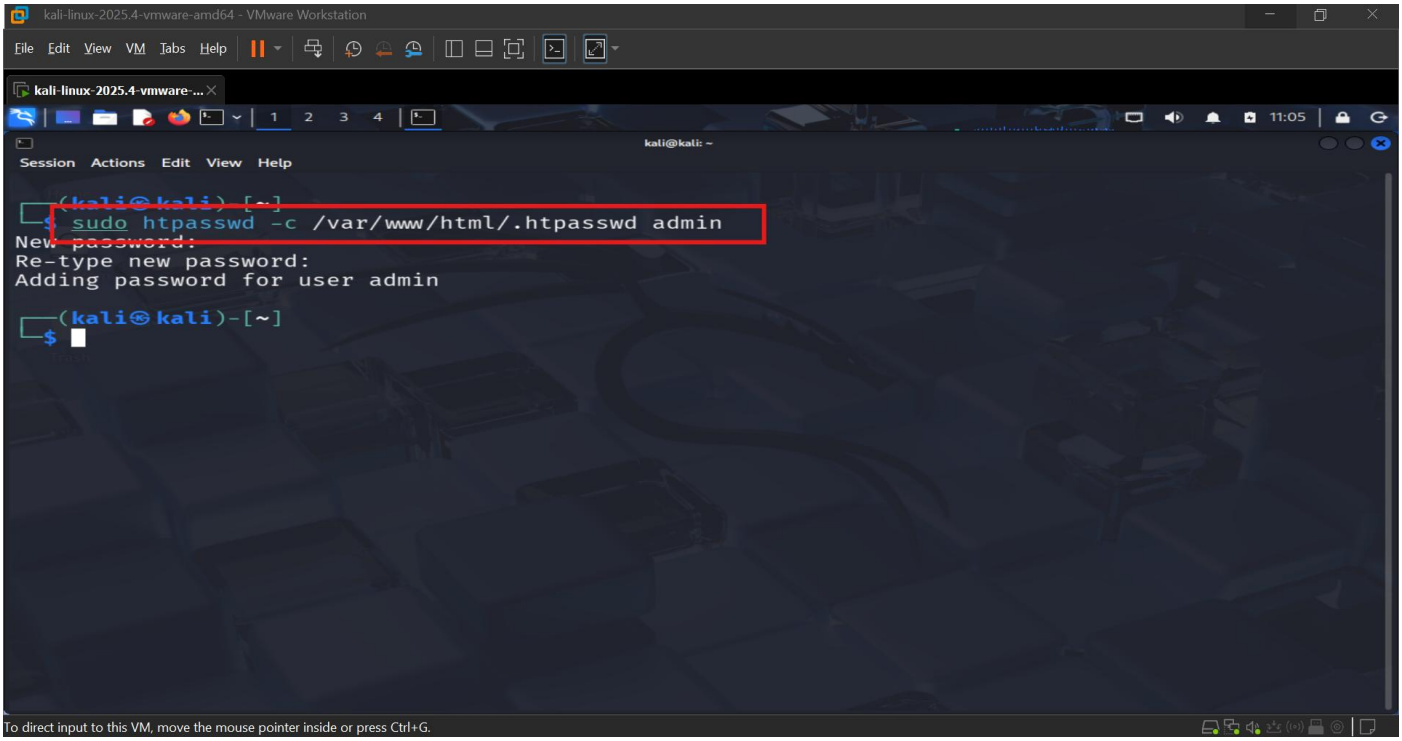
5.5 Secure Access by editing .htaccess
Create .htaccess file in /var/www/html:
Command: **sudo nano /var/www/html/.htaccess**



Add or modify the following lines:
AuthType Basic
AuthName "Restricted Access"
AuthUserFile /var/www/html/.htpasswd
Require valid-user



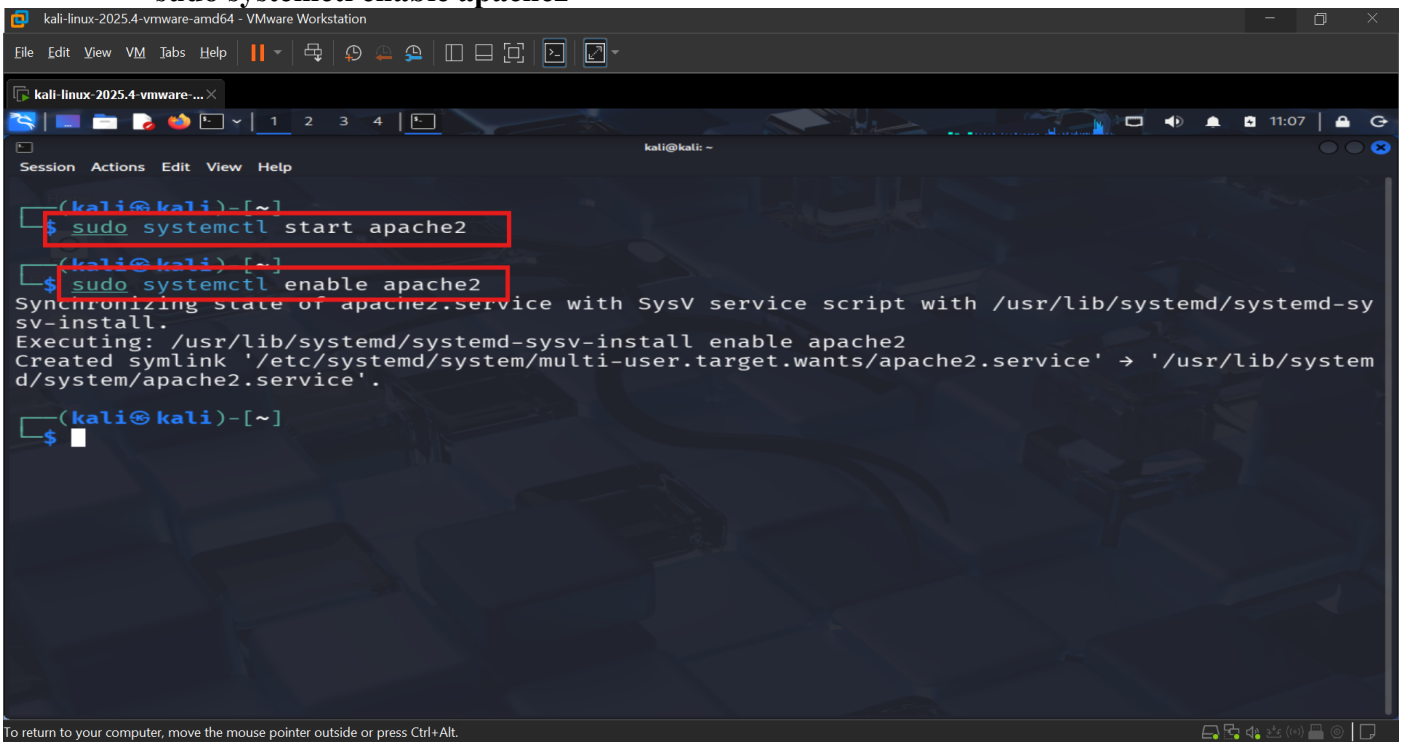
5.6 Create .htpasswd File for Authentication
Command: **sudo htpasswd -c /var/www/html/.htpasswd admin**



5.7 Start and enable apache2

Command: **sudo systemctl start apache2**

sudo systemctl enable apache2



Step 6: Testing and Validation

To confirm that your configuration is properly set up, you should conduct several tests to validate that permissions, command history settings, and logging are functioning as expected.

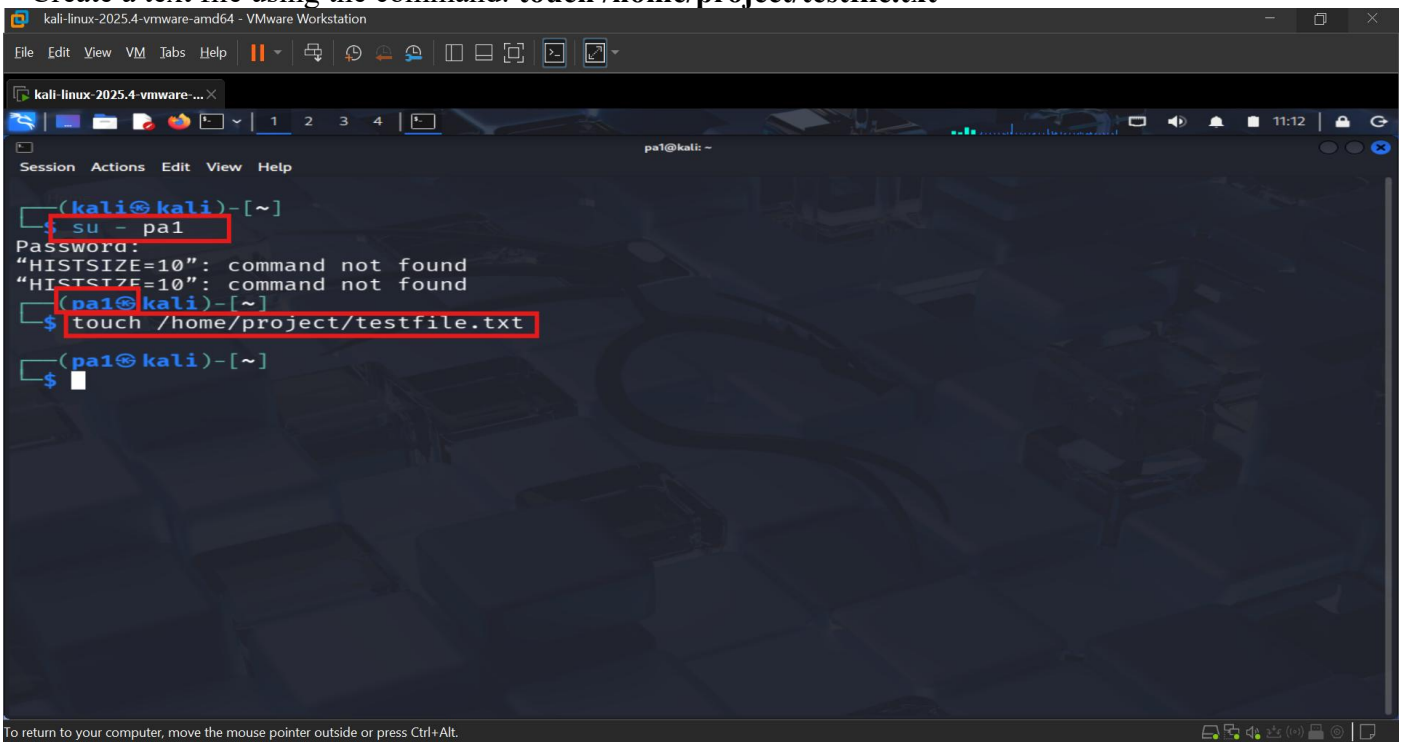
6.1: Test File Creation and Modification:

Log in as each user (pa1, pa2, pa3, pa4, pa5, pb1, pb2, pb3) and try creating a file in the /home/project directory.

Example:

- Log in as a **pa1** user using the command: **su - pa1**

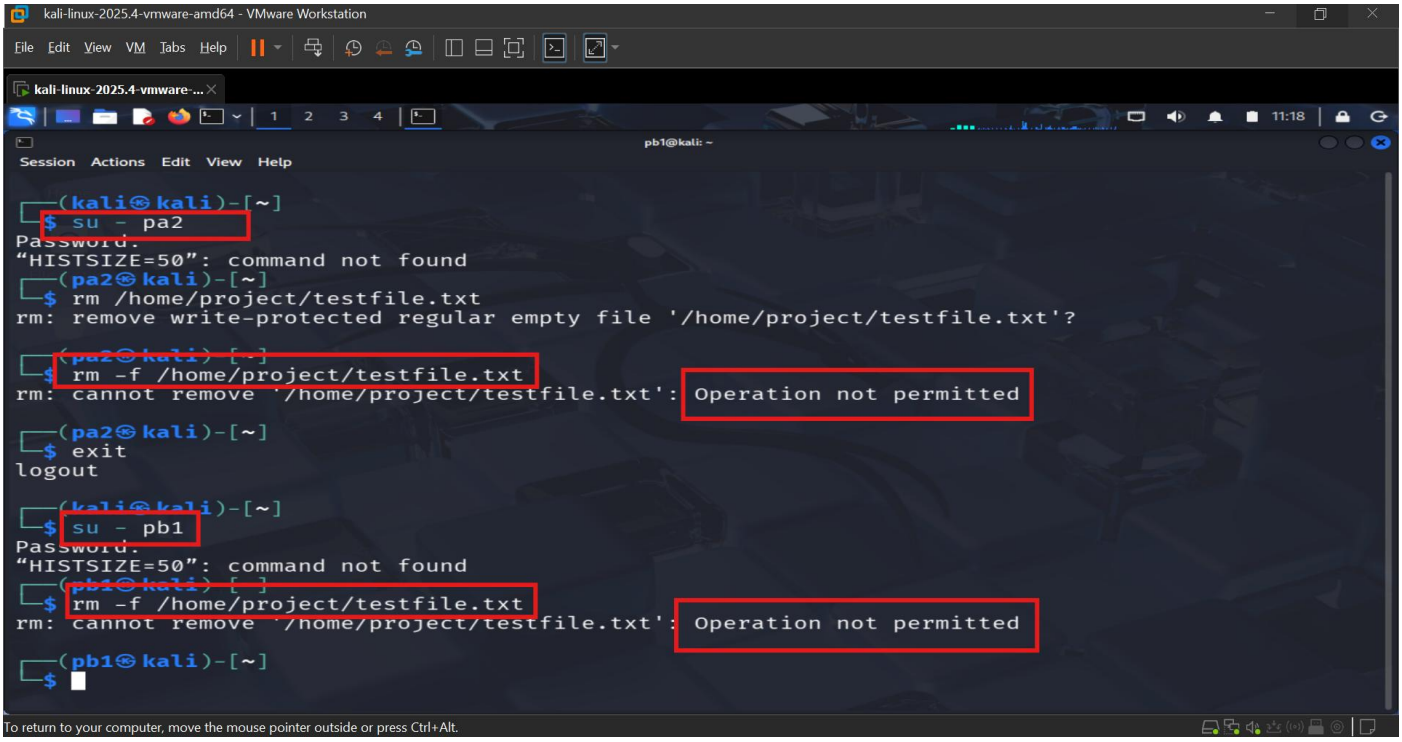
● Create a text file using the command: **touch /home/project/testfile.txt**



Now try to remove the created file through another user.

● First login as any user using the command: **su - pa2**

● Remove the text file using the command: **rm -f /home/project/testfile.txt**

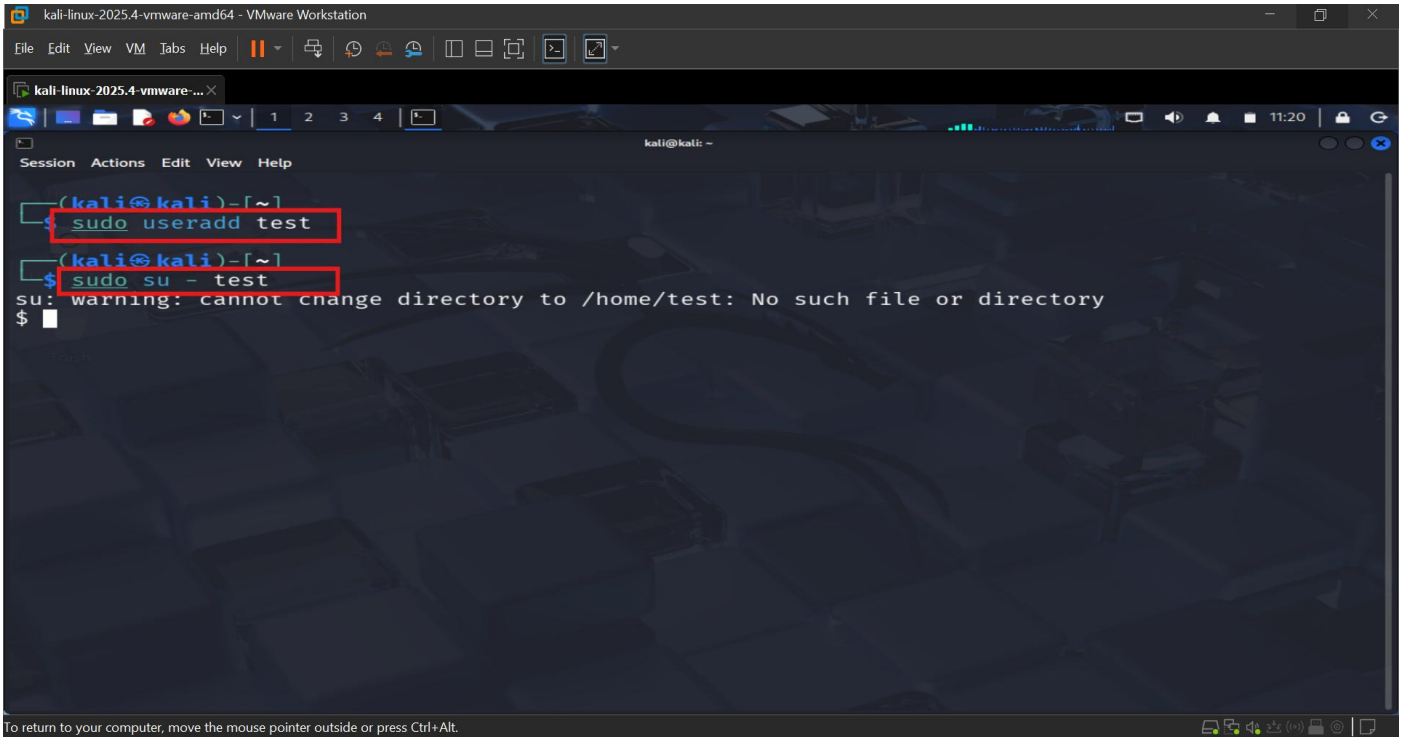


6.2 Unauthorised Access Attempt:

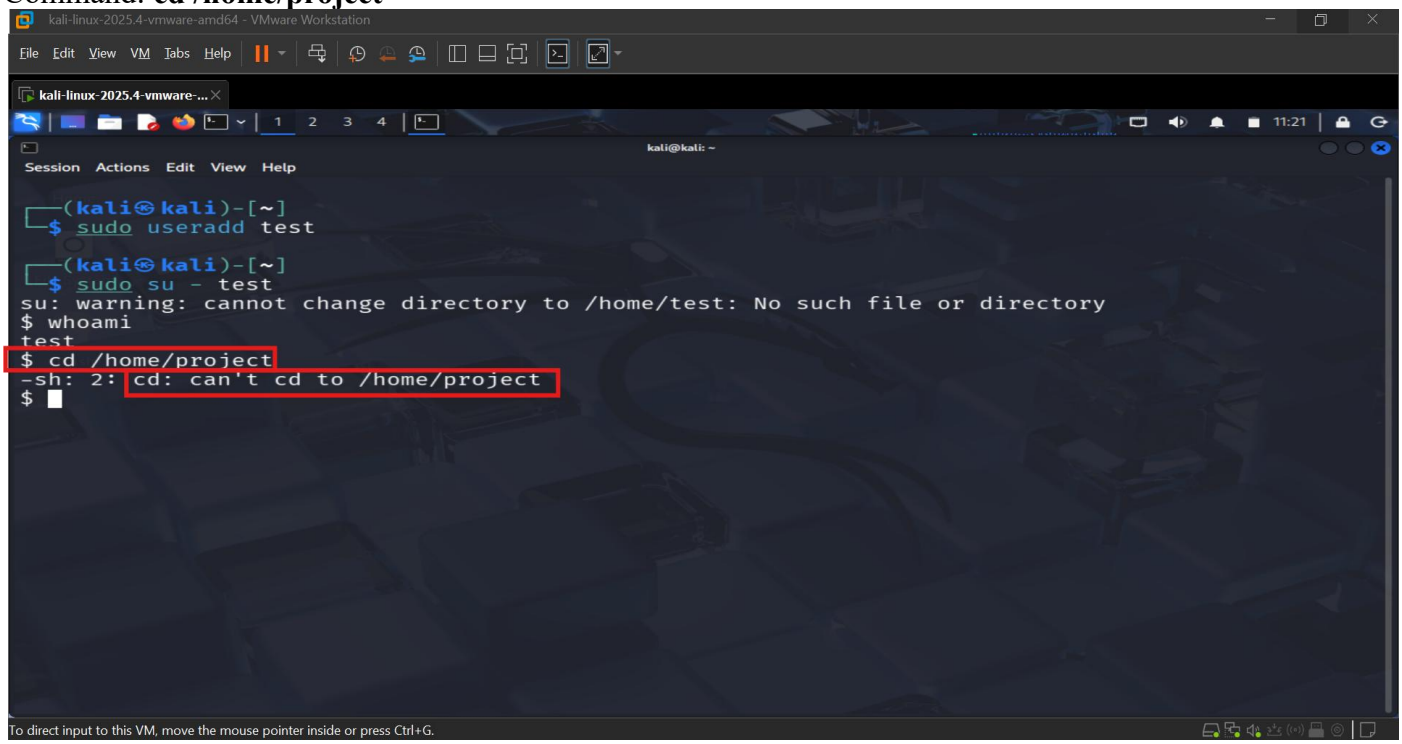
● Create a test user test and try accessing the /home/project directory. We are assuming that the "test" user already exists in the system.

● Command: **sudo useradd test**

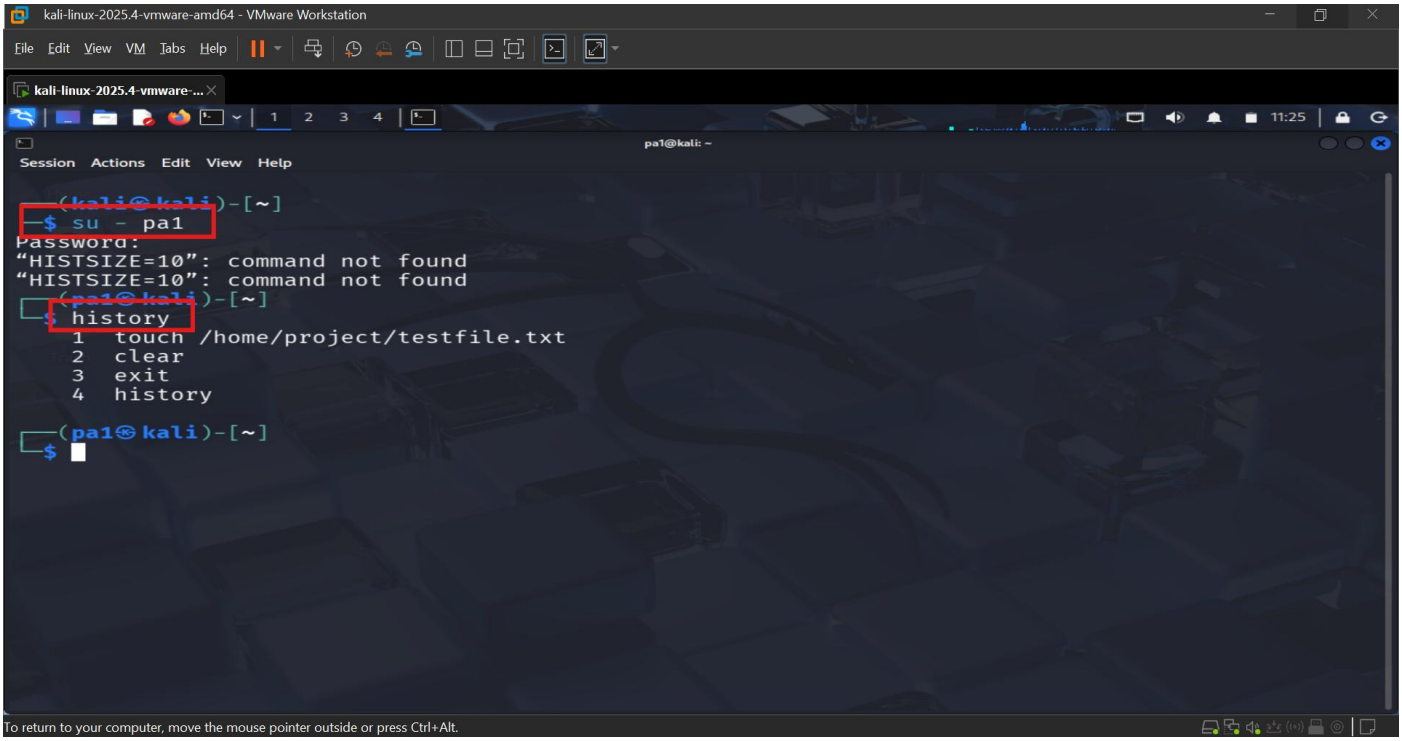
sudo su - test



Now try to change the directory
Command: **cd /home/project**

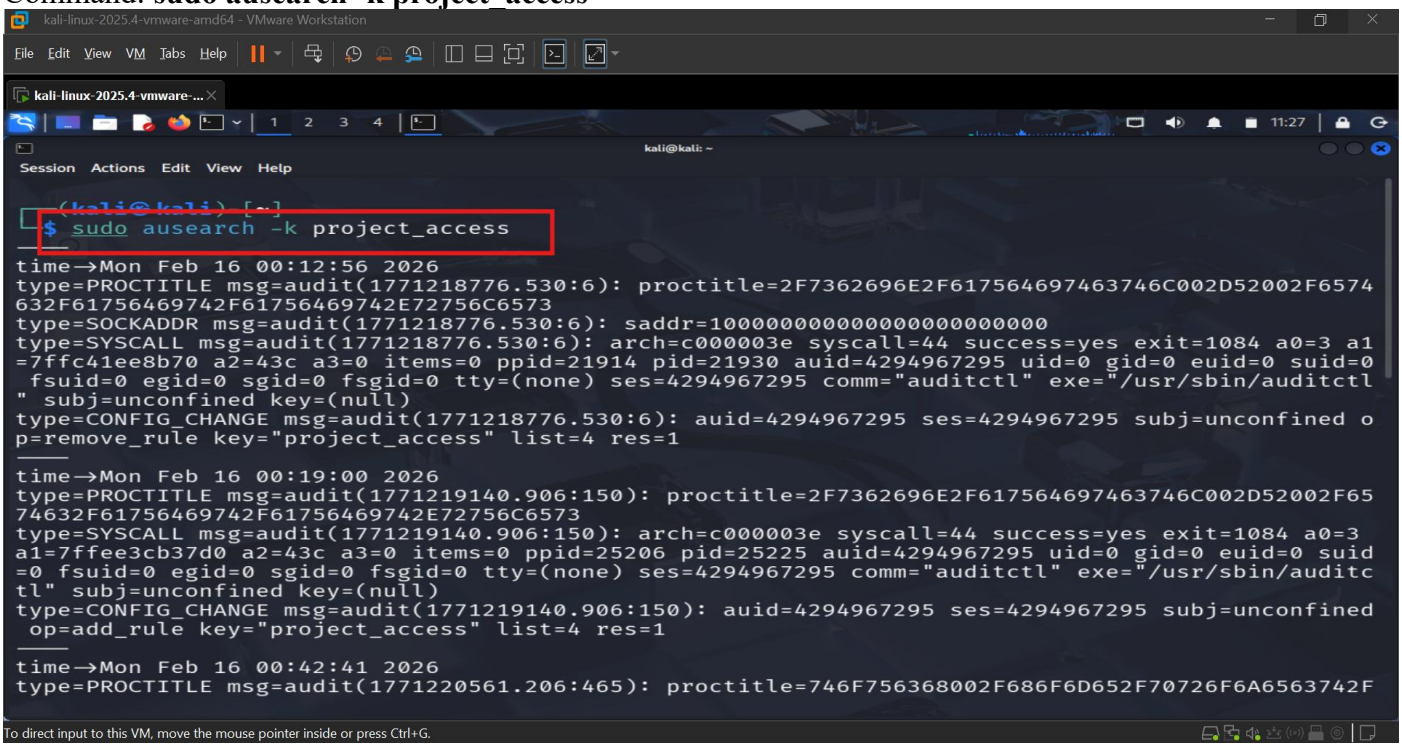


6.3 Testing Command History Configuration
To validate that the command history settings are correctly applied for each user:
- Log in as pa1 or pa5:
- Use command: **history**
- The output should display the last 10 commands.



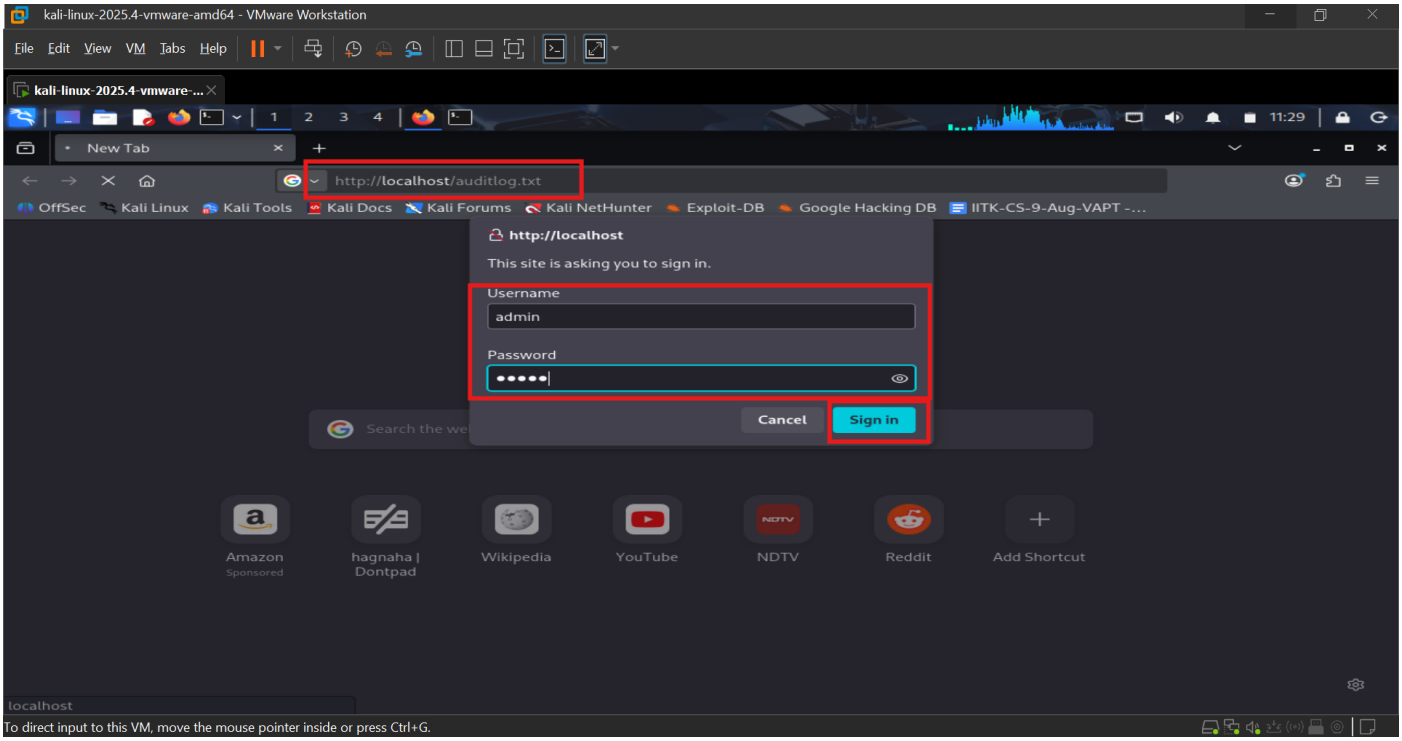
6.4 Testing Audit Logging of Unauthorised Access Attempts

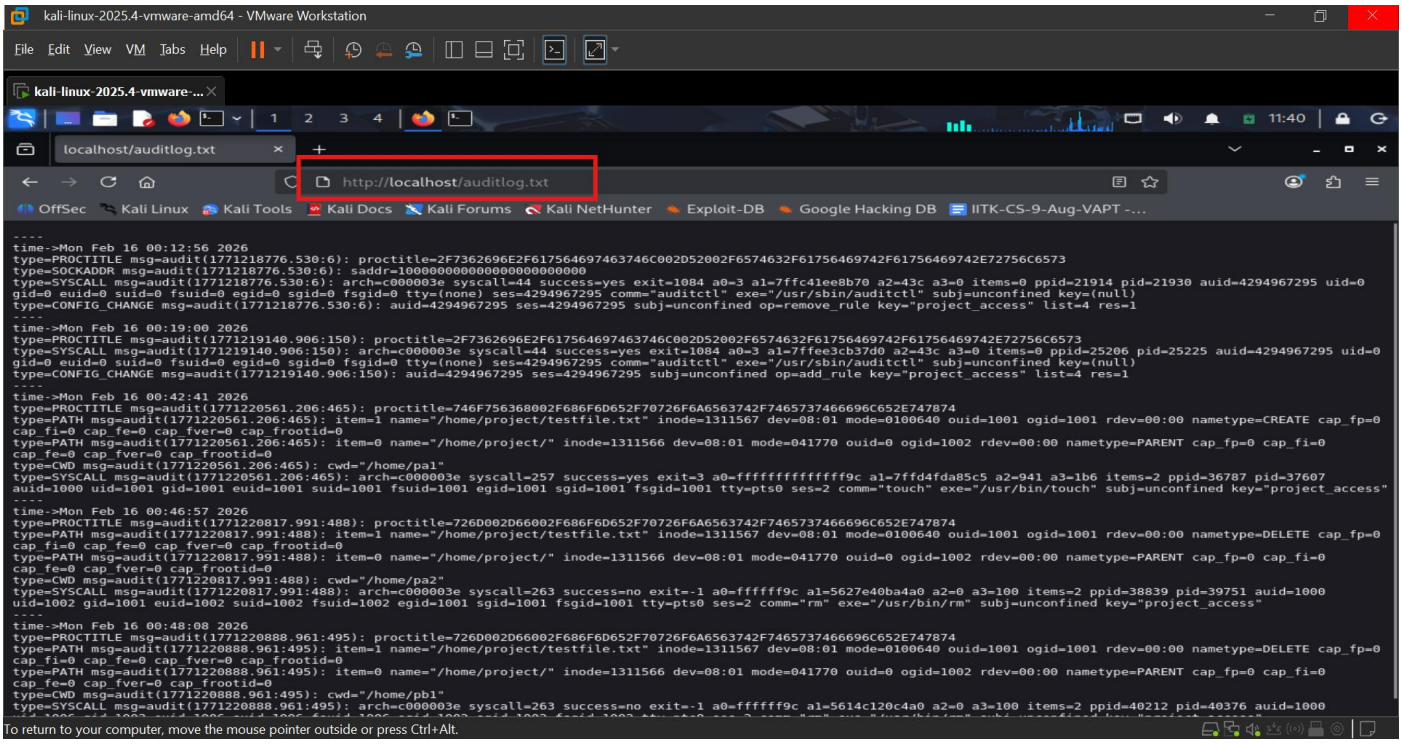
Command: **sudo ausearch -k project_access**



6.5 Test the Web-Based Report System

Open a web browser and navigate to the URL of your web-based report system (e.g., **http://localhost/auditlog.txt** or **http://ip-address/auditlog.txt**).





Conclusion

This project demonstrates how **Linux security mechanisms** such as user/group management, ACLs, auditd, and access logging can be combined to build a **robust enterprise-level file security solution**.

By implementing strict access controls, monitoring user activity, and maintaining audit logs, the system ensures **confidentiality, integrity, and accountability** of sensitive project data.

The project highlights the importance of proactive security controls in organizational environments and provides a scalable foundation for future enhancements.

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